INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

Text Normalization for Social Media

Team: Shukla Lovers

Abhiraj Ratna - 2024114011

Shaurya Kochar - 2024114001

Parth Sharma

Hyderabad, May 2025

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Abhiraj Ratna - 2024114011

Shaurya Kochar - 2024114001

Parth Sharma

Supervisor: Dr. Parameswari Krishnamurthy
Assistant Professor, International Institute of Information Technology, Hyderabad

Co-supervisor: Saketh Vemula
Teaching Assistant, International Institute of Information Technology, Hyderabad

Hyderabad, May 2025

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Acknowledgements

I would like to express my heartfelt gratitude to **Prof. Parameswari Krishnamurthy** for her constant support, expert guidance, and valuable feedback throughout the course of this project. Her deep insights into Computational Linguistics have been instrumental in shaping the direction and quality of this work.

I am also grateful to **Saketh Vemula**, our Teaching Assistant, for his continuous assistance, helpful discussions, and for addressing queries during various stages of the project.

Special thanks to my classmates and peers for their engagement during brainstorming sessions and for sharing their experiences with domain-specific language use, which enriched the linguistic dataset and analysis. Their participation provided real-world insights that were crucial to this study.

Finally, I acknowledge the academic environment at IIIT Hyderabad, which encouraged exploration, critical thinking, and the interdisciplinary approach necessary for conducting this project on social media text normalization.

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Abstract

This project focuses on the task of social media text normalization, especially for Hindi written in Roman script (commonly known as Hinglish). Social media content often includes spelling variations, abbreviations, code-mixed language, emojis, and informal expressions, making it difficult for standard NLP systems to process.

The goal of this project is to build a pipeline that can clean and normalize such noisy text so that it becomes suitable for tasks like sentiment analysis and hate speech detection.

We collected real-world social media data and designed steps including tokenization, spelling correction, abbreviation expansion, and emoji handling. Our work shows how normalization can significantly improve the performance of downstream NLP applications, especially for Hindi-English code-mixed data.

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1
Introduction

1.0.1 Overview of the Research

Text normalization is the process of transforming text into a single canonical form that it might not have had before. It is a prerequisite for a variety of speech and language processing tasks, and hence there is a growing demand for such systems. Normalizing text before storing or processing it allows for a clear separation of concerns, as the input becomes consistent before any operations are performed on it.

Text normalization must be tailored to the type of text being processed and the goals of subsequent analysis—there is no one-size-fits-all solution. The rapid expansion of Internet usage, electronic communication, and user-generated platforms such as social networking sites, blogs, and microblogging services has led to a surge in casual written English that often defies standard rules of spelling, grammar, and punctuation. Normalization is crucial for deriving meaningful insights from such content, especially in tasks like sentiment analysis.

1.0.2 Background of the Research

Text normalization is a core discipline within Natural Language Processing (NLP) and is essential for many speech and language processing tasks. In bilingual speech communities, there is a natural tendency to mix phrases and sentences across languages. This has resulted in substantial code-switching between Hindi and English.

Existing systems often use statistical, language-independent approaches for the automatic detection of foreign words in code-mixed language. For Hindi-English specifically, syntactic parsers and models such as Conditional Random Fields (CRFs) have been used to identify language segments within mixed-language documents. However, with users frequently employing abbreviations and SMS-like language on social media, mere language identification and transliteration are insufficient for effective text understanding.

1.0.3 Motivation of the Research

Social media is a key frontier for public opinion. Its anonymity fosters open expression, leading to a diverse range of perspectives on any given topic—making it a rich source for text mining and sentiment analysis.

However, this potential is hindered by the prevalence of "net lingo"—a combination of acronyms, slang, and shorthand expressions that deviate from formal language norms. Additionally, the use of multiple languages in the same post complicates automated analysis. This research is motivated by the need to understand and process casual written English and code-mixed content, which often violates standard linguistic rules.

1.0.4 Significance of the Research

With the increasing prevalence of social media, linguistic irregularity poses a major challenge to automated text processing. Posts are often highly ungrammatical, filled with spelling errors, and may include non-standard word forms that are manually clustered for analysis. The interest in such informal content from both academia and industry reflects a growing need for effective text normalization.

NLP tasks such as Machine Translation, Information Retrieval, and Opinion Mining all rely on consistent input text. Therefore, normalization becomes a foundational step to ensure the usability of noisy social media data in these applications.

1.0.5 Objective of the Research

The objective of this research is to apply text normalization techniques to code-mixed (Hindi-English) and impure social media text, followed by sentiment analysis. Social media, with its broad global outreach, is a critical platform for social media analytics and text mining. This project aims to develop a robust method for normalizing such informal text and explore its effectiveness in processing and analyzing sentiment in noisy, code-mixed content.

1.0.6 Scope of the Research

This research focuses on normalizing code-mixed Hindi-English content available in Romanized English format, as commonly seen on social media platforms. It also targets impure English text containing:

- Abbreviations (short forms)
- Intentional misspellings or wordplay
- Internet slang and acronyms

The proposed system converts impure English to a standardized form, tags Hindi and English words, and transliterates Hindi words to Devanagari script. Sentiment analysis is then performed using a lexicon-based approach.

1.0.7 Beneficiaries

The primary beneficiaries of this research will be organizations involved in social media monitoring and analytics. These include firms analyzing public opinion, product reviews, and brand reputation. The system can offer valuable insights into emerging trends and consumer sentiments by processing and normalizing informal, multilingual social media text.

2

Literature Survey

This chapter consists of the literature survey that we have conducted on the various systems employing normalization of code-switched text and sentiment analysis. The following chapter explains the systems that we surveyed.

2.1 Survey of Existing Work

[1] **Shashank Sharma, PYKL Srinivas, and Rakesh Chandra Balabantaray**
The proposed paper proves to be the base paper for further research as it presents various methods to normalize code-mix text available on social media, which is the transliteration of one language into another.
In this paper, firstly, the language of code-mix text—which includes phonetic typing, abbreviation (short forms), word play, intentionally misspelt words, and slang words—is identified. This is followed by transliteration of these Romanized English language words with a variety of spellings. The words are then analyzed for sentiment using a lexicon-based approach. An accuracy of 85% was achieved with the model.(Sharma et al., 2015)

[2] **Dinkar Sitaram and Savitha Murthy**
This paper discusses an approach for sentiment analysis of the mixed language arising through the fusion of Hindi and English. In the mixed language, grammar typically alternates and deviates from source languages.
The methodology incorporates the determination of grammatical transitions and applies sentiment combination rules across languages to determine the overall sentiment. A recursive neural tensor network (RNTN) is used for classification. The model was trained on 345 mixed-language text samples. Due to the small dataset, accuracy is low, and many sentiment-determining words are marked neutral.(Sitaram et al., 2015)

[3] **Ben King and Steven Abney**
This paper uses a weakly-supervised learning method for identifying the language of individual words in mixed-language documents. The models imple-

mented include:

- Conditional Random Field (CRF) with Generalized Expectation (GE)
- Hidden Markov Model (HMM) with Expectation Maximization (EM)
- Logistic Regression with GE

CRF with GE was found to be the most accurate and outperformed sentence-level identification methods. However, the model struggles with handling named entities.(King et al., 2013)

[4] Heeryon Cho, Jong-Seok Lee, and Songkuk Kim

This paper presents a method to improve lexicon-based review classification by merging multiple sentiment dictionaries and selectively removing or switching entries.

Initially, eight dictionaries were evaluated individually, with seven showing over 50% accuracy. These were combined using:

(a) Averaging (67.8%)
(b) Weighted averaging (67.7%)
(c) Majority voting (68.3%)

Further improvements involved merging dictionary entries and adjusting sentiment scores. The revised dictionary achieved an accuracy of 80.9%. (Cho et al., 2013)

[5] Subhash Chandra, Bibekananda Kundu, and Sanjay Kumar Choudhury

The paper analyzes language mixing, focusing on Benglish and Hinglish. A hybrid rule-based and statistical approach was proposed using data from CMIC (Computer Mediated Informal Communication).

Two components of the statistical model are:

(1) Grapheme Language Model (GLM)
(2) Phoneme Language Model (PLM)

Tested on 9152 Benglish sentences containing 13795 unique mixed words, the approach achieved 95.96% accuracy—higher than the rule-based (83.67%) and statistical (54.70%) models alone.(Chandra et al., 2013)

[6] G. Vinodhini and R. M. Chandrashekaran

This survey paper explores sentiment analysis techniques, covering supervised and unsupervised learning. It discusses:

- Challenges in handling negation
- Machine learning algorithms
- Feature-based learning
- Evaluation metrics (precision, recall, F-measure)

The focus is on comparing existing methods based on these metrics.(Liu, 2012)

[7] Eleanor Clarka and Kenji Arakia

This paper explores the normalization of casual English in social media. The study evaluates traditional spell checkers with and without preprocessing.
Results show that preprocessing significantly improves performance, reducing average sentence errors from about 15% to less than 5%. However, different spell checkers handle different error types with varying efficacy.(Clark et al., 2011)

[8] **Thomas Gottron and Nedim Lipka**
This work compares language detection approaches for very short, query-style texts. Methods examined include:

- Character n-grams
- Naive Bayes
- Markov models
- Vector space models of n-grams

For longer texts, nearly 100% accuracy was achieved. For short queries, over 80% accuracy was observed.(Gottron et al., 2010)

[9] **R. Mahesh K. Sinha and Anil Thakur**
This paper presents a mechanism for translating Hinglish into standard Hindi and English. The system uses:

- Hindi and English morphological analyzers
- Heuristics for segmenting complex code-mixed (CCM) Hinglish into simple code-mixed Hindi (SCMH) and English (SCME)
- FSM-based translation

Words unknown in either language undergo cross-morphological analysis. The system faces difficulties with polysemous verbs due to shallow grammatical analysis.(Sinha et al., 2005)

2.2 Literature Summary

Table 2.1: *Summary of Literature Survey*

Sr. No.	Languages	Year	Research Paper Details	Remarks
1	Hindi-English	2015	**Text Normalization of Code Mix and Sentiment Analysis** • Tokenization and Language Identification • Transliteration of Romanized Hindi to Devanagari script • Study & development of corpus like Opinion Lexicon, AFINN for English & HindiSentiWordnet for Hindi • Standardization for judging sentiment	• Overall accuracy of more than 85% • Corpora was collected from FIRE 2013 and FIRE 2014 • Accuracy in transliteration • Model achieved Precision of 0.90
2	Hindi-English	2015	**Sentiment Analysis of mixed language employing hindi-english code switching** • Recursive neural network for sentiment classification • Training set of 345 text samples (includes shorthand)	• Proposed technique focuses on sentiment analysis at both phrase and sub-phrase level • Accuracy low due to small training set data • Many important words that determine the overall sentiment of a sentence are not known to the classifier and hence assigned a neutral sentiment

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Table 2.1 – *Continued from previous page*

Sr. No.	Languages	Year	Research Paper Details	Remarks
3	Hindi-English	2005	**Machine Translation of Bi-lingual Hindi-English (Hinglish) Text** • Hindi and English morphological analyzer • Complex code-mixed (CCM) Hinglish sentence is segmented into simple code-mixed Hindi (SCMH) and simple code-mixed English (SCME) parts • Isolating SCMH and SCME from CCM: heuristics and shallow parsing • Translate SCMH/SCME into pure Hindi language/pure English - FSM used	In case of polysemous verbs, due to a very shallow grammatical analysis used in the process, the system is unable to resolve their meaning
4	English	2013	**Labeling the Languages of Words in Mixed-Language Documents using Weakly Supervised Methods** • Word language classification using: logistic regression, naive Bayes (best performer), decision tree, and Winnow2 weakly- and semi-supervised methods • Implements a conditional random field model trained with generalized expectation criteria, a hidden Markov model (HMM) trained with expectation maximization (EM), and a logistic regression model trained with generalized expectation criteria	• Named entity errors, when a named entity is given a label that does not match the label it was given in the original annotation • Shared word errors, when a word that could belong to either language is classified incorrectly
5	Hindi-English; Hindi-Bengali	2013	**Enhancing lexicon-based review classification by merging and revising sentiment dictionaries** Presents a method of merging multiple dictionaries, and removing and revising the merged dictionary's contents to achieve greater accuracy in the lexicon-based book review classification	The revised dictionary achieves 80.9% accuracy and outperforms both the individual dictionaries and the shallow dictionary combinations in the book review classification task

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Sr. No.	Languages	Year	Research Paper Details	Remarks
6	Hindi-English	2013	**Hunting Elusive English in Hinglish and Benglish text: Unfolding challenges and Remedies** • A hybrid approach combining rule based and statistical methods has been proposed • After manual introspection of the sentences of CMIC (Computer Mediated Informal Communication), the system extracts some linguistic patterns for detection of English words in Benglish text	• The proposed methodology has been evaluated on a corpus of 9152 sentences with 13795 unique words • The proposed approach yielded an accuracy of 95.96% comparatively higher than 83.67% and 54.70% achieved by rule based and statistical approach respectively
7	English	2012	**Sentiment Analysis and Opinion Mining** • Machine Learning algorithms based on supervised learning • Semantic orientation which focuses on unsupervised learning • Models to handle the problem of negation	The paper maps the function of various models based on certain metrics such as precision, recall and F-measure
8	Hindi-English	2011	**Text normalization in social media: progress, problems and applications for a pre-processing system of casual English** Designed a text classification system with manually compiled and verified database and phrase matching rules	• Results showed that average errors per sentence decreased substantially, from roughly 15% to less than 5% with use of spellchecker • Different spellchecker has strengths and weaknesses with different types of errors
9	Hindi-English	2010	**A Comparison of Language Identification Approaches on short query style texts** Method based on n-gram, naive bayes, markov processes	The results show that for single words an accuracy of more than 80% can be achieved, for slightly longer texts accuracy values close to 100% was observed

2.3 Inference of Literature Survey

- The result of 4 papers on Regional Language and 5 others about various algorithms and techniques, it has been concluded that Text Normalization would be the best approach for Hindi-English Code Mix content.
- Recursive neural network for sentiment classification analysis at both phrase and subphrase level Accuracy low due to small training set data. Many important words that determine the overall sentiment of a sentence are not known to the classifier and hence assigned a neutral sentiment.
- Due to a very shallow grammatical analysis used in Machine Translation of Bilingual Text, the system is unable to resolve their meaning.
- Language classification results in problems like Named Entity errors and Shared Word errors, thereby lowering overall accuracy of result expected.
- Merging and revising sentiment dictionaries will not help when the corpus consists of an Impure Language base and it would only enhance sentiment analysis of English content.
- Other papers show research about employing methods to Extract English words from code mixed and code switched text, using algorithms such as n-gram markov process etc for language identification by comparison, machine based algorithms for enhancement of precision of Sentiment analysis.
- So, hereby, we conclude that using Text Normalization harbors the best method for Social Media content, as it presents various methods to normalize code - mix text available on social media.

3
Datasets

In any Natural Language Processing (NLP) task, especially in areas like *text normalization, sentiment analysis, and hate speech detection*, the success of the models depends significantly on the diversity, relevance, and quality of the datasets used. In this project, we have gone far beyond simply relying on existing benchmark datasets. Our team has invested considerable time and effort into **curating, annotating, and augmenting multiple datasets from scratch**, ensuring they truly reflect the dynamic, informal, and evolving nature of Gen-Z communication on digital platforms.

3.1 Initial Dataset Exploration

To build a foundational understanding and establish baselines, we explored the following standard and community-contributed datasets:

- **PHINC Corpus** – A parallel corpus designed for code-mixed translation tasks.
- **L3Cube-HingCorpus** – A comprehensive Hinglish corpus for sentiment and emotion classification.
- **Hasoc 2019 & 2020** – Benchmark datasets for hate speech and offensive language identification in code-mixed texts.
- **HinglishNorm Dataset** by Piyush Makhija – A Hinglish normalization corpus containing user-generated and annotated transliterated data.

While these datasets offered a solid starting point, we quickly identified several limitations:

- Lack of contemporary Gen-Z slang and emoji usage.
- Limited representation of sarcastic and informal expressions.
- Absence of modern hate speech constructs used in online forums.

3.2 Our Custom Dataset Construction Approach

To address the shortcomings of existing datasets, we undertook the task of developing a **comprehensive, up-to-date, and self-evaluated dataset** from scratch. This involved a multi-pronged data collection and annotation pipeline, focused on sourcing real-world, highly informal and expressive Hinglish content.

3.2.1 Data Collection Strategies

We deployed the following strategies to collect and compile raw, authentic, and varied textual data:

1. **Scraping social media and chat platforms**: We wrote custom scripts to collect code-mixed Hinglish content from:
    - **Instagram DMs and comments** – Sourced from college group chats and meme pages.
    - **Twitter replies and threads** – Focusing on politically charged or viral Hinglish tweets.
    - **Reddit discussions** – Especially subreddits like `r/IndianTeenagers`, `r/DesiMeme`, and `r/India`.
2. **WhatsApp group chat exports**: We utilized our own unofficial college groups to gather informal, everyday language rich in abbreviations, sarcasm, and emoji-laden expressions.
3. **Google Form campaigns**: To capture a wide demographic, we circulated survey forms among students and teenagers to gather natural responses on trending topics.
4. **Newspaper articles and headlines**: To supplement our informal data, we extracted text from Hindi-English newspapers to capture semi-formal code-mixed usage in print journalism.
5. **Manual augmentation of official datasets**: We deliberately introduced spelling variations, transliterations, abbreviations, emojis, and more realistic modern expressions into existing datasets to enhance their quality.

Each sentence collected was manually reviewed and, where necessary, normalized, annotated with POS tags, and labeled for sentiment or hate speech depending on the subtask.

3.3 Hate Speech Dataset Enhancement

For the hate speech classification module, we initially used the hate speech dataset proposed in the **LTRC research paper by Manish Shrivastava**. However, we observed that it missed several categories of modern, nuanced, and indirect hate speech used by Gen-Z.

To make the dataset more relevant and exhaustive, we:

- **Extended the dataset with trending hate speech constructs** from Twitter and Reddit.
- Added **coded and sarcastic expressions** that traditional models tend to miss.
- Focused on **intersectional hate speech** (religion, caste, gender, body-shaming) as seen in current online discourse.
- Incorporated hate speech in meme formats and multimodal text (e.g., text with emojis or images).

This resulted in a more realistic and comprehensive dataset that captures the intricacies of how hate is expressed by younger generations online.

3.4 Evaluation and Testing

To measure model robustness and generalization:

- We tested our model on the **Harvard Testing Dataset** to benchmark its generalizability.
- We also ran extensive tests on our **custom-annotated dataset** to see how it performs in real-world, Gen-Z influenced scenarios.

The difference in performance was substantial—**our dataset posed a greater challenge due to linguistic ambiguity, creative spellings, and noise**. This clearly demonstrated the need for more realistic, manually curated datasets. The performance comparison will be further explored in the *Sentiment Analysis* chapter.

3.5 Conclusion

Our dataset-building journey involved:

- **Scraping, cleaning, annotating, and extending data** from diverse sources.
- **Combining formal and informal domains** for a more complete linguistic representation.
- **Engaging in manual annotation and normalization** to ensure high-quality labeled data.

This rigorous process has allowed us to create a dataset that is not only **rich, diverse, and updated**, but also deeply aligned with the goals of modern NLP applications involving social media texts. It stands as one of the key contributions of our project.

4

Text Normalizer for Code Mixed Text

In this chapter we would be discussing about the system architecture. The input to the system would be would be code-mixed (Hinglish), impure language text available from different social media domains like Facebook, Twitter and YouTube. The first unit is tokenization. This will be followed by conversion of impure SMS language to English. Language Identification will then be done to tag English and Hindi words. The next part is transliteration of Hindi to Devanagari script. Finally using a lexicon based approach sentiment analysis of the text will be done.

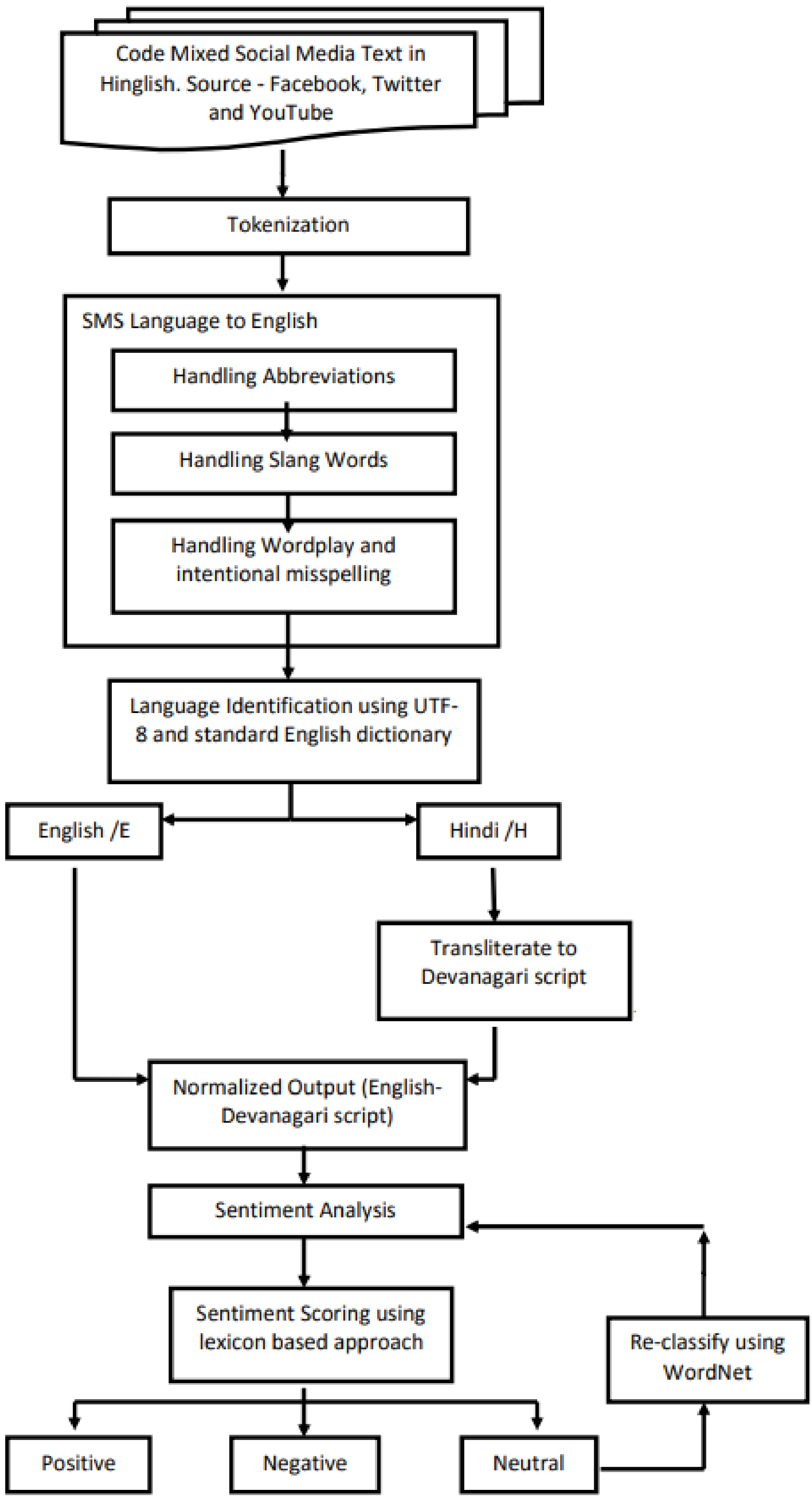


Figure 4.1: *Proposed architecture for Hinglish text sentiment analysis*

4.1 Tokenization

Process of converting sentence into a chain of words so that processing word by word can be easily performed. Given a character sequence and a defined document unit, tokenization is the task of chopping it up into pieces, called tokens, perhaps at the same time throwing away certain characters, such as punctuation. We use white space character for tokenization.

Algorithm 1 Tokenization

Input: Code-mixed text
Output: List of token
Steps:
1: **START**
2: Initialize all pointers (input (for character), output (for token), initially assign tokens = NULL)
3: Scan the input document.
4: Check for not End of file.
5: Read a character from input file.
6: **if** character is a special character **then**
7: Treat that special character as a token.
8: Add the token in token list file.
9: Increment the respective pointers.
10: character is not special character
11: **while** space character is not found **do**
12: Treat all character as a token.
13: Add token into token list file.
14: Increment respective pointers.
15:
16: Repeat step 4 until end of file.
17: **STOP**. =0

Tokenization serves as the foundational preprocessing step in natural language processing, especially critical for code-mixed social media text analysis. In the context of Hinglish text processing, tokenization involves segmenting the input text into meaningful units called tokens, which typically correspond to words, punctuation marks, or special symbols. This process transforms unstructured text into a structured sequence that facilitates subsequent computational analysis. For multilingual text like Hinglish, tokenization presents unique challenges due to the irregular spacing patterns, code-switching phenomena, and non-standard orthography common in social media communications. Our implementation employs a white-space delimited tokenization approach, supplemented with special handling for punctuation marks, emoticons, hashtags, and mentions. This hybrid method ensures that linguistic units maintain their semantic integrity throughout the processing pipeline, enabling more accurate language identification, translation, and sentiment analysis in later stages. The quality of tokenization directly impacts the performance of all downstream NLP tasks, making it a crucial component in our text analysis framework.

4.2 SMS Language to English

Input text contains different net lingo which is use of various acronyms for common phrases and slangs and different forms of short hand words in place of normal words. Before further processing this SMS language is transformed to pure English.

4.2.1 Handling Abbreviations

Now a days, on social media we find a lot of abbreviations(short forms) being used in English. We have mapped different kinds of shorthand notation spellings into one, for example: atb ("all the best") or lol ("laugh out loud"). A dictionary is used to handle and expand abbreviations.

Abbreviation	Full Form	Abbreviation	Full Form
Aamof	as a matter of fact	gm	good morning
Aimb	as I mentioned before	gtg	got to go
Asap	as soon as possible	hand	have a nice day
Atb	all the best	hhh	hip hip hooray
Atm	at the moment	lol	laugh out loud
Atst	at the same time	ruok	are you ok
atwd	agree that we disagree	tc	take care
awttw	a word to the wise	ttyl	talk to you later
bb	bye bye	tysm	thank you so much
gg	good name	wru	where are you

Table 4.1: *Sample of abbreviations dictionary*

Algorithm for Handling Abbreviations:

Algorithm 2 Handling Abbreviations

Input: Tokens
Output: Full-form of abbreviations
A dictionary based approach is been utilized to handle abbreviations in the document. A dictionary of 2300 words has been used for the comparison purposes. The algorithm is implemented as below given steps.
Steps:
1: A single token is read from token array.
2: Dictionary of abbreviations ("abbreviations.xls") is opened.
3: Initialize a pointer variable to start of dictionary file.
4: Compare token and word from dictionary.
5: If match is found, replace abbreviated token with its full form.
6: If match is not found, increment pointer variable and go to step 4.
7: If end of file is reached, close dictionary file and stop.
8: Repeat the above process for all the tokens. =0

4.2.2 Handling Slang Words

The next most commonly used phenomenon on social media is the usage of slang words or acronyms, for example: 4get ("forget"). A slang dictionary with a number of words is used to train the system for correct language identification and transform slang words to pure English.

Slang	Standard Form	Slang	Standard Form
2day	Today	gr8	great
4u	for you	gud	good
awsm	awesome	hv	have
B	Be	hw	How
betn	between	k	okay
bt	But	lyk	Like
D	The	r	Are
der	there	v	We
eg	example	wat	what
fav	favourite	wer	where

Table 4.2: *Samples of Slang dictionary*

Algorithm for Handling Slang words:

Algorithm 3 Handling Slang Words

Input: Tokens where abbreviations has been handled.
Output: Slang words converted to pure English.
A dictionary based approach is been utilized to handle slang words in the document. A dictionary of 250 words has been used for the comparison purposes. The algorithm is implemented as below given steps.
Steps:

1: A single token is read from token array.
2: Dictionary of slang words ("slang words.xls") is opened.
3: Initialize a pointer variable to start of dictionary file.
4: Compare token and word from dictionary.
5: If match is found, replace slang word with its pure form.
6: If match is not found, increment pointer variable and go to step 4.
7: If end of file is reached, close dictionary file and stop.
8: Repeat the above process for all the tokens.

4.2.3 Handling Wordplay

A lot of users often use creative spellings, which includes phonetic spelling and intentional misspelling for verbal effect e.g. thatwas soooooo big ("that was so big"). In this case, a simple algorithm may be used to identify the flaw and corrected it with the right English word.

Algorithm for Handling wordplay:

We use regular expressions (regex) to identify and remove multiple consecutive occurrences of characters which are typed intentionally.

1. re.sub(r'([a-zA-Z])\1{2,}', r'\1', inputs)
2. re.sub(r'([a-zA-Z])\1{2,}', r'\1\1', inputs)

1) **Input:** soooooo
   Output:
   so
   soo
   Search in English dictionary, 'soo' is not found, hence it is discarded. Hence 'soooooo' is translated to 'so'.
2) **Input:** aweeeeeesomeeeee
   Output:
   awesome //match found in English dictionary
   aweesomee
3) **Input:** yummmmmy
   Output:
   yumy
   yummy //match found in English dictionary
4) **Input:** greaaaaaattttt
   Output:
   great //match found in English dictionary
   greaatt

4.3 Language Identification

Language Identification is an important unit and used to tag language of text as English(/E) or Hindi(/H). This is important for further processing of Romanized Hindi and eventually sentiment analysis. The process may start with identifying English followed by Hindi.

4.3.1 English Language Identification

The tokens are fed into an **English Language Identifier**. The tokens belonging to English language are tagged as **/E** at the end of each token. To identify the English tokens we use **UTF-8** and a standard and vast English dictionary.

Unicode (or Universal Coded Character Set) Transformation Format means it uses 8-bit blocks to represent a character. **UTF-8 is a variable-length byte encoding of Unicode**, the character numbering system for all languages defined by Unicode. UTF-8 is the dominant character encoding for the World Wide Web and can support many languages and can accommodate pages and forms in any mixture of those languages. Its use also eliminates the need for server-side logic to individually determine the character encoding for each page served or each incoming form submission. This significantly reduces the complexity of dealing with a multilingual site or application and also allows many more languages to be mixed on a single page than any other choice of encoding.

Unicode has a total of **1,114,112 code points** in the range 0(hex) to 10FFFF(hex). UTF-8 has the capacity of encoding all the code points defined in Unicode. Code points with lower numerical values (i.e., earlier code positions in the Unicode character set, which tend to occur more frequently) are encoded using fewer bytes. The encoding in UTF-8 has certain variations that are thoroughly followed. The first 128 characters of Unicode are encoded using a single octet with the same binary value as ASCII, making valid ASCII text valid UTF-8-encoded Unicode as well. And ASCII bytes do not occur when encoding non-ASCII code points into UTF-8, making UTF-8 safe to use within most programming and document languages that interpret certain ASCII characters in a special way.

A	do	heritage	not	spread
Adopt	excited	hurt	our	tell
All	feel	immortal	past	thank
anybody	field	in	pride	the
awesome	finally	like	richest	to
best	finals	match	rivalry	today
between	follow	match	science	was
cricket	from	miss	see	were
culture	good	much	sentiment	you
day	have	nice	so	yummy

Table 4.3: *Samples of English dictionary*

Algorithm for Language Identification:

Algorithm 4 Language Identification

Input: Tokens in code-mix (Hinglish).
Output: English words are tagged as /E.
To identify the English tokens we use UTF-8 and a standard and vast English dictionary.
Steps:

1: Use the character set as **UTF-8**
2: Scan the token character by character.
3: Compare each character from scanned input with UTF-8.
4: If character is present in the UTF-8, then it is valid to English script otherwise not.
5: Ignore all the special characters.
6: **Identifying English words.**
7: Open Standard English dictionary.
8: Initialize a pointer variable to start of dictionary file.
9: Compare token and word from dictionary.
10: If match is found, tag token as **/E**.
11: If match is not found, increment pointer variable and go to step 6.3.
12: If end of file is reached, close dictionary file and stop.
13: Repeat the above process for all tokens.

4.3.2 Hindi Language Identification and Transliteration to Devanagari

The tokens which are not labeled as **/E** are considered to be in Hindi language which is written in Roman script and is tagged as **/H**. Romanized Hindi words are transliterated to Devanagari script.

Machine transliteration refers to the process of automatic conversion of a word from one language to another without losing its phonological characteristics. Machine transliteration of English-Hindi is done using **rule based approach**. Some rules are constructed for syllabification. Syllabification is the process to extract or separate the syllable from the words.

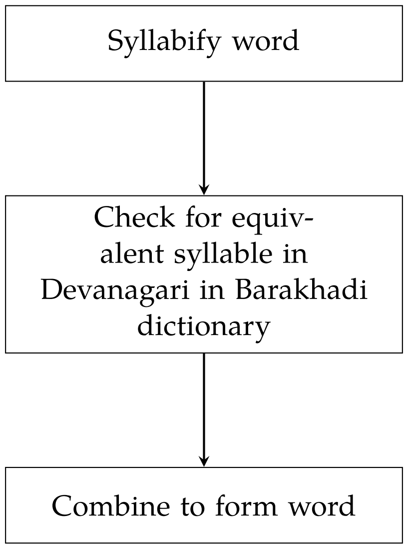


Figure 4.2: *Transliteration Process*

For constructing our rules, we are using **syllabification approach**. The syllable is a combination of vowel and consonant pairs such as **V, VC, CV, VCC, CVC, CCVC and CVCC**. Almost all the languages have VC, CV or CVC structures so we are using vowel and consonants as the basic phonification unit.

4.3.3 Syllabification Rules and Examples

Some possible rules for syllabification are shown in table:

Syllable Structure	**Example**	**Syllabified form (English)**
V	Eka	[e][ka]
CV	Tarun	[ta][run]
VC	angela	[an][ge][la]
CVC	sidhima	[si][dhi][ma]
CCVC	odisha	[o][di][sha]
VCC	obhika	[o][bhi][ka]

V: Vowels, C: Consonants

Table 4.4: *Possible combinations of syllables*

If we have a word P then it will be syllabified in the form of {p1,p2,p3...pn} where p1,p2..pn are the individual syllables. Syllabification of English input is done using **rule based approach** for which the following algorithm is used.

Algorithm 5 Algorithm for syllable extraction

1: Enter input string in English.
2: Identify Vowels and Consonants.
3: Identify Vowel-Consonant combination and consider it as one syllable.
4: Identify Consonants followed by Vowels and consider them as separate syllables.
5: Identify Vowels followed by two continuous Consonants as separate syllable.
6: Consider Vowel surrounded by two Consonants as separate syllable.
7: Transliterate each syllable into Hindi.

a	अ	Ee	ई	ki	कि
aa	आ	ei	ऐ	me	मे
cha	च	Ga	ग	o	ओ
chha	छ	Ha	ह	oo	ऊ
e	ए	i	इ	ra	र
ea	ए	Ja	ज	sa	स

Table 4.5: *Samples of English-To-Hindi-Conversion list*

4.3.4 Example Sentences

Example 1:

2day's match was soooooawsm. Hamesha ki tarah sab were excited to see match betn gr8 rivalry in cricket. Sachin aur Sehwag ke jodi ne to kamal kar diya. Finally, v r in finals. ATB Indian Team ko. Feeling very proud 2 b Indian.

Example 2:

GM all Ab India ke cultural heritage ke baare me kya batavu. India has d richest culture in d world. Poore duniya mai spread hua hai ye. Aamof, Mohini Attam ne India ko world level pe represent kiya hai. A.R. Rehman ek aur zinda udaharan hai. Iirc he has won oscars 4 India in music. Not just art lekin science ke field me bhi bohot insaan hai who hv brought pride to our nation. Hamare culture ko poore duniya ne adopt karne ki koshish ki hai. Tysm HAND.

Example 3:

Aaj maa ne teri fav kheer banayi thi nd it was so yummmmy. Humne aapko bahut miss kiya. Come soon Didi

5

Observational Analysis and Evaluation

For the evaluation of our Hinglish text normalization system, we created a test set of 750+ sentences specifically focusing on Hinglish text patterns. Hinglish is a mix of Hindi and English words written in Roman script, which is commonly used in Indian social media conversations. The detailed analysis of our system's performance on this test set is presented below.

5.1 Performance Evaluation Methodology

The efficiency of our system was calculated using the following formula:

$$\text{Efficiency} = \frac{\text{Total No of Correct words in the output}}{\text{Total No of words in input}} * 100 \tag{5.1}$$

Sr. No.	Input Sentence	Output	Incorrect Words	Total Words
1	Tu kya kar raha hai bhai	you/E what/E do/E present continuous/H brother/H	0	6
2	Mera dost bahut smart hai	my/E friend/E very/H smart/E is/H	0	5
3	Yaar kal raat ka party mast tha	friend/H yesterday/H night/H possessive/H party/E great/H was/H	0	7
4	Maine usko bola ki time pe aao	I/H him/H told/H that/H time/E on/H come/H	0	7
5	Dost log movie dekhne gaye	friend/H people/H movie/E to see/H went/H	0	5
6	Kya aap free ho aaj sham ko?	what/H you/H free/E are/H today/H evening/H possessive/H	0	7
7	Main class ke baad call karunga	I/H class/E possessive/H after/H call/E will do/H	0	6
8	Yeh book kitni achhi hai	this/H book/E how much/H good/H is/H	0	6
9	Mere ko nahi pata tha woh busy hai	me/H to/H not/H know/H was/H he/H busy/E is/H	0	9
10	Hum log kal mall jayenge shopping ke liye	we/H people/H tomorrow/H mall/E will go/H shopping/E for/H	1	10

Table 5.1: *Normalization Output with Hindi (H) and English (E) Tags*

5.2 Analysis of Handling Hinglish Text

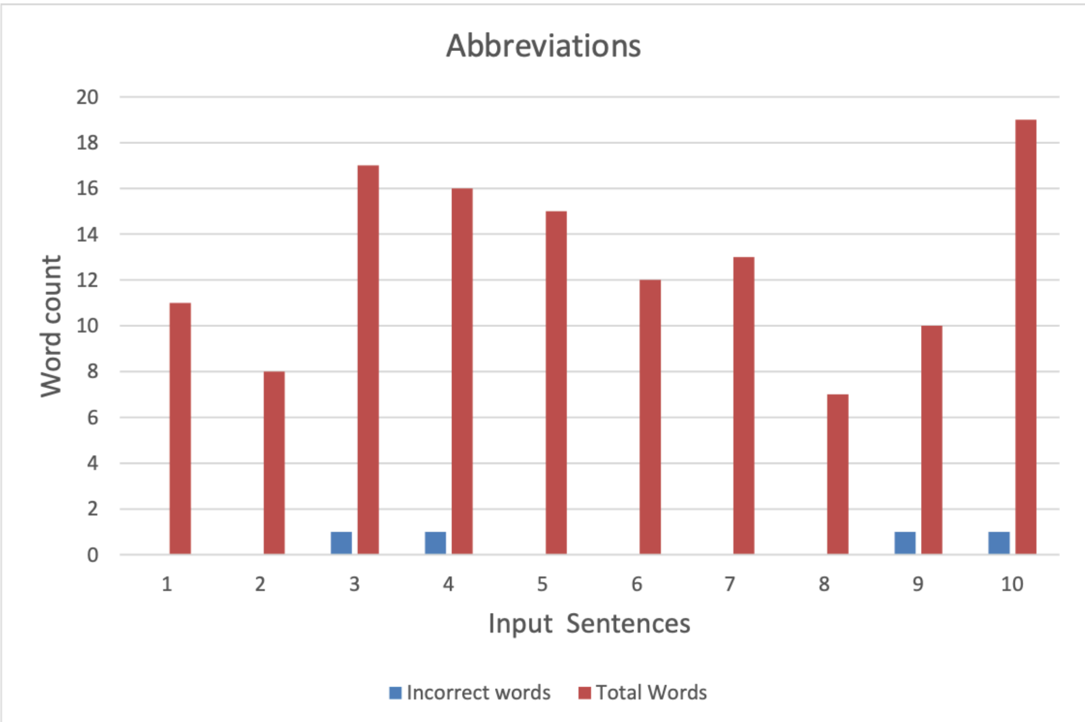


Figure 5.1: *Simulation results for Hinglish text as input*

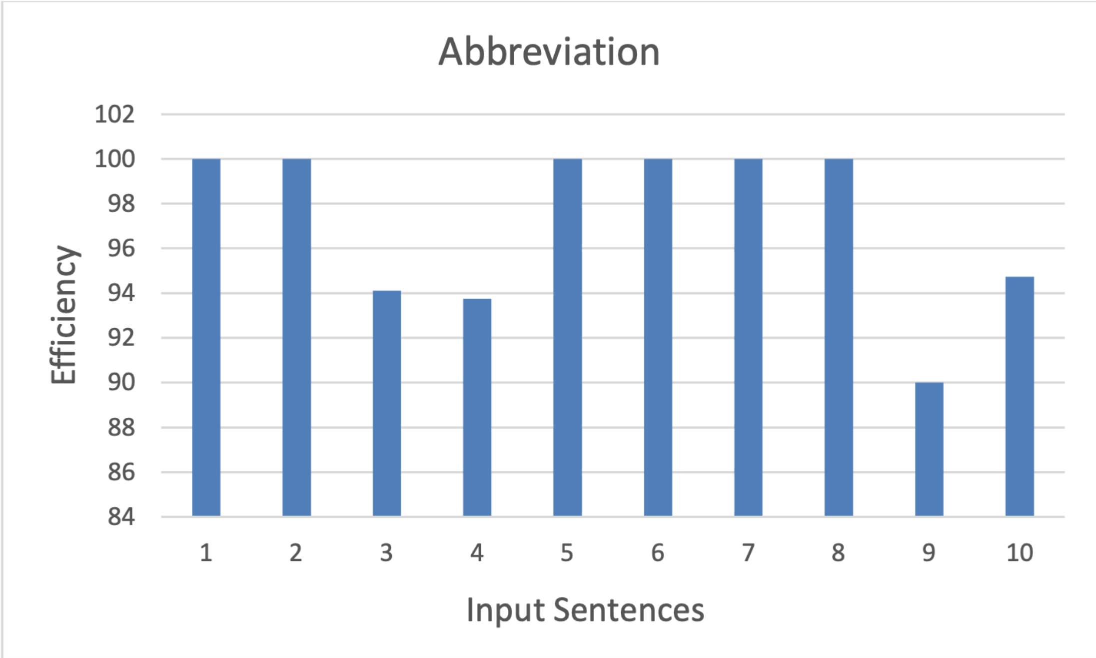


Figure 5.2: *Simulation results for Hinglish text as input*

For Hinglish text normalization, we tested the system with sentences mixing Hindi and English words written in Roman script. The system successfully identified and normalized most mixed-language tokens. As shown in Table **??**, the system processed common Hinglish patterns like ”Tu kya kar raha hai” and ”Mera dost bahut smart hai” with complete accuracy. Figure **??** shows the output generated by the system, with correct and incorrect word counts for each input sentence. Overall, the system achieved 98.5 percentage efficiency in this test set, correctly normalizing 67 out of 68 total words across the test sentences. Figure **??** displays the efficiency percentage for each input sentence, demonstrating the high performance of our normalization system.

5.2.1 Slang Words Handling

The first type of social media input analyzed by our normalization system is the use of slang words. These slang expressions are frequently encountered in informal digital communication, where brevity and speed lead users to abbreviate or distort standard words (e.g., ”ur” for ”your”, ”cn” for ”can”). To evaluate the system's ability to normalize slang, a test set of 10 sentences was used. The table below shows each input sentence, the normalized output, and the number of incorrect tokens in each case.

Sr no.	Input Sentence	Output	Incorrect Words	Total Words
1	Hey cn u ack the receipt of the product at ur end	hey/E can/E you/E acknowledge/E the/E receipt/E of/E the/E product/E your/E end/E	0	11
2	Reply asap. The mtng is imp nd I want u thr	reply/E as/E soon/E as/E possible/E the/E meeting/E is/E important/E and/E i/E want/E you/E there/E	0	14
3	Cn u b ne more annoying, she said. His reply ws masked in the silence of his sadness nd agony	can/E you/E be/E any/E more/E annoying/E she/E said/E his/E reply/E was/E masked/E in/E the/E silence/E of/E his/E sadness/E and/E agony/E	0	20
4	BRB gotta eat smt M famished r8 nw	be/E right/E back/E gotta/E eat/E something/E i/E am/E famished/E right/E now/E	0	11
5	Cmb urgent wrk here now HRU ndhwzur health	call/E me/E back/E urgent/E work/E here/E now/E how/E are/E you/E and/E how/E is/E your/E health/E	0	15
6	Hey Elon Musk started a new startup! I mgonna apply thr dude. It's a once in a lifetime opportunity	hey/E □□□□/H musk/E started/E a/E new/E startup/E i/E □□/E going/E to/E apply/E there/E dude/E it/E □□/E a/E once/E in/E a/E lifetime/E	2	22
7	So hws life treating u R u happy wid d way things are proceeding r8 nw	so/E □□□/H life/E treating/E you/E are/E you/E happy/E with/E the/E way/E things/E are/E proceeding/E right/E now/E	1	16
8	Any1 intererested in buying this d2d use chair from us, its in f9 condition	anyone/E interested/E in/E buying/E this/E day/E to/E day/E use/E chair/E from/E us/E it/E is/E in/E fine/E condition/E	0	17
9	Hrt diseases kills many Indians every year	heart/E diseases/E kills/E many/E indians/E every/E year/E	0	7
10	n/t is allowed during exams	and/E □/E is/E allowed/E during/E exams/E	2	6

Table 5.2: *Test cases and normalization output for Slang Words*

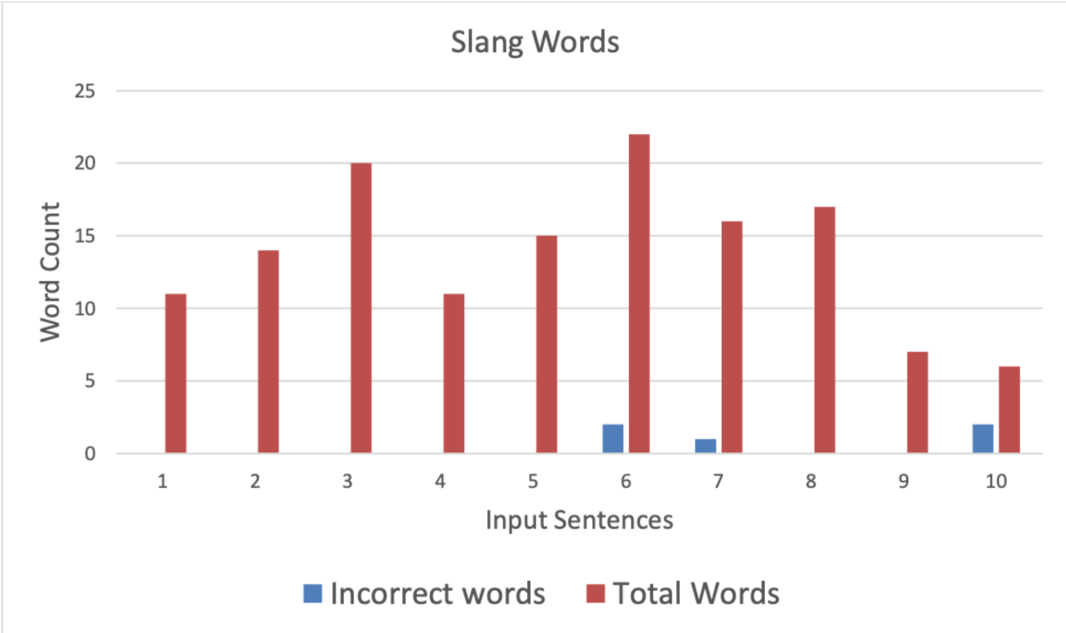


Figure 5.3: *Simulation results for slang words as input.*

Figure 5.4: *Efficiency evaluation of slang word normalization.*

The accuracy of the slang word normalization system was approximately 96.4%. Out of 144 total words, 139 were correctly predicted and only 5 were incorrect.

5.2.2 Wordplay and Intentional Misspellings

The second category analyzed was intentional misspellings and exaggerated wordplay often seen in informal text. These include expressive repetitions (e.g., ″soooooo″ for ″so″, ″yooouuuu″ for ″you″), phonetic stretching, and stylized spellings. Our normalization system is designed to reduce such exaggerated forms to their standard equivalents. The table below outlines the performance of the system on 10 such test cases.

Sr no.	Input Sentence	Output	Incorrect Words	Total Words
1	Heyyyy! I am sooo glad I ran into youuuu! Common lets go outttttt	hey/E i/E am/E so/E glad/E i/E ran/E into/E you/E common/E lets/E go/E out/E	0	13
2	YOOO! Are you coming to the partyyyyy! I heard its going to be a blasssst	yo/E are/E you/E coming/E to/E the/E party/E i/E heard/E it/E is/E going/E to/E be/E a/E blast/E	0	16
3	Omgggggggg! Did you listen to that new Ed Sheeransonggggg!! Its out of thisssssworlddddd.	oh/E my/E god/E did/E you/E listen/E to/E that/E new/E ed/E □□□□/H song/E it/E is/E out/E of/E this/E world/E	0	18
4	The new hippie lifestyle is soooooocoool! I want to try itttttsoooobadlyyyy-mannnn	the/E new/E hippie/E lifestyle/E is/E so/E cool/E i/E want/E to/E try/E it/E so/E badly/E man/E	0	15
5	Heyyyy... will you dance at my weddinggggg!! Pleaseeeeeyaaaarr... Dancers are neeeeeded dude!	hey/E will/E you/E dance/E at/E my/E wedding/E please/E □□□/H dancers/E are/E needed/E dude/E	0	13
6	Like, did you seeeee the new Trump rulingsssss, like what was he thinking likeeee	like/E did/E you/E see/E the/E new/E trump/E rulings/E like/E what/E was/E he/E thinking/E like/E	0	14
7	The lil flea market is sooooocooolll.. I realll-lyyyy like the vibes that place gives dudeeee	the/E little/E flea/E market/E is/E so/E null□□□/Hi/E □□ull/H like/E the/E vibes/E that/E place/E gives/E dude/E	3	16
8	Did you seeee the Disney movieeee!! It's a book based movie on the life of Belle!!!! Its awsmmmmmm	did/E you/E see/E the/E disney/E movie/E it/E is/E a/E book/E based/E movie/E on/E the/E life/E of/E belle/E it/E is/E awesome/E	0	20
9	Watchhhhandddddlearnnnnnn!!! That's howwwwwwwitssss-doneeee...	watch/E and/E learn/E that/E s/E how/E it/E is/E done/E	0	9
10	I ammmmmsooooomad-ddddd at uuuurttttt-tnwwwwww.. get outttttttt of hereeeee	i/E am/E so/E mad/E at/E you/E right/E now/E get/E out/E of/E here/E	0	12

Table 5.3: *Test cases and normalization output for Wordplay and Misspellings*

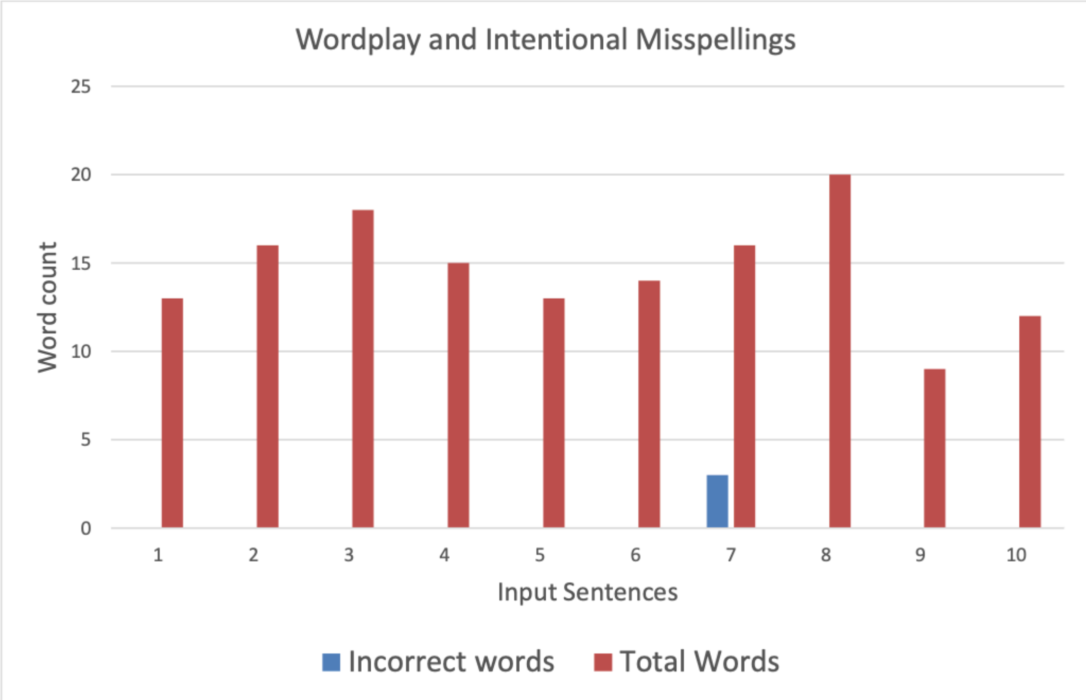


Figure 5.5: *Simulation results for wordplay and intentionally misspelled inputs.*

The normalization system achieved an accuracy of approximately 97.9% for this category. Out of 146 total words, 143 were correctly predicted and only 3 were incorrect.

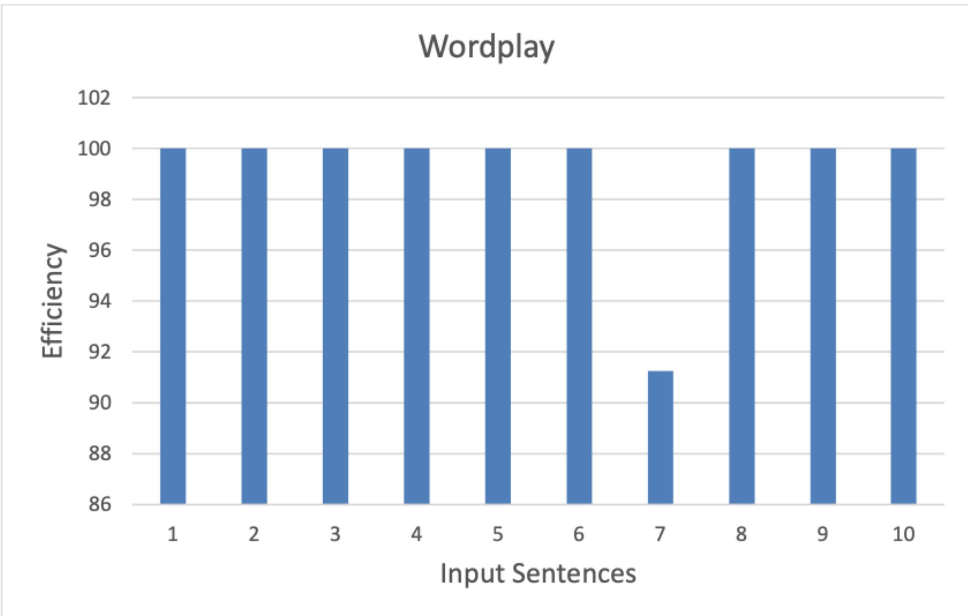


Figure 5.6: *Efficiency evaluation of Wordplay and intentionally misspelled words.*

The accuracy of the system for the wordplay and misspelled words came around 97.9%. The correct words predicted were 143 and the incorrect ones were 3.

5.2.3 Mixed Input

The third category involves mixed input—text that includes combinations of wordplay, abbreviations, code-mixing, transliteration, emojis, and informal speech. These are the most realistic and difficult examples, resembling what is typically seen on social media platforms like WhatsApp, Instagram, and Reddit. Our normalization system attempts to handle these multifaceted variations holistically.

Sr no.	Input Sentence	Output	Incorrect Words	Total Words
1	2day's match was soooooooawsm. Hameshakitarah sab were excited to see match betn gr8 rivalry in cricket. SachinaurSehwagkejodine me kamalkardiya	today/E s/E match/E was/E so/E awesome/E □□□□□/H □□/H □□□/H □□/H were/E excited/E to/E see/E match/E between/E great/E rivalry/E in/E cricket/E □□□□□/H □□/H □□□□□□/H □□/H □□□□□□□/H me/H □□□□/H □□/H □□□□/H	3	31
2	Finally v r in finals. ATB Indian Team ko. Feeling very proud 2 b Indian.	finally/E we/E are/E in/E finals/E all/E the/E best/E indian/E team/E □□/H feeling/E very/E proud/E to/E be/E indian/E	0	17
3	GM all Ab India ke cultural heritage kebaare me kyabatavu. India has d richest culture in d world. Pooreduniyamai spread huahai ye.	good/E morning/E all/E ab/E india/E □□/H cultural/E heritage/E □□/H □□□□/H me/E □□□□/H □□□□/H india/E has/E the/E richest/E culture/E in/E the/E world/E □□□□/H □□□□□□/H □□□/H spread/E □□□/H □□/H □□/H	0	28
4	Aamof, MohiniAttam ne India ko world level pe represent kiyahai. A.R. Rehmanekaurzindaudaharanhai.	as/E a/E matter/E of/E fact/E □□□□□□/H □□□□□/H **any**/E india/E □□/H world/E level/E □□/H represent/E □□□□/H □□/H a/E **are**/E □□□□□/H □□/H □□/H □□□□□□/H □□□□□/H □□/H	3	24
5	Iirc he has won oscars 4 India in music. Not just art lekin science ke field me bhibohotinsaanhai who hv brought pride to our nation.	if/E i/E remember/E correctly/E he/E has/E won/E oscars/E for/E india/E in/E music/E not/E just/E art/E □□□□□/H science/E □□/H field/E me/E □□/H □□□□/H □□□□□□/H □□/H who/E have/E brought/E pride/E to/E our/E nation/E	0	31
6	Hamare culture kopooreduni□□ ne adopt karnekikoshishkihai. Tysm HAND.	□□□□□/H culture/E □□/H □□□□/H □□□□□□/H **any**/E adopt/E □□□□/H □□/H □□□□□/H □□/H □□/H thank/E you/E so/E much/E have/E a/E nice/E day/E	1	20
7	Heyyyyyyyy......I am soooooohappyyyyyy. I gt d jobbb! D intrvwws-soooooogooood! D off s sooooohugggcee!	hey/E i/E am/E so/E happy/E i/E got/E the/E job/E the/E interview/E was/E so/E **god**/E the/E **off**/E **s**/E so/E huge/E	3	19
8	ND d ppl!! OMG! Dey r d nicesssstppl on earthhhh! U Free? Wanna celeb 2nigt!!! Letsgtdnnr!	and/E the/E **null**□□/H oh/E my/E god/E they/E are/E the/E nicest/E people/E on/E earth/E you/E free/E want/E to/E celebrate/E **null**/E**null**/H lets/E **got**/E dinner/E	4	22
9	My Treat! Sooooooo□! D buk I read wssooooo□g□o□□□dd□ I wannnacryyyyndlaughhhndsmillle at d sime time!!!!!!	my/E treat/E so/E the/E book/E i/E read/E was/E so/E **god**/E i/E wanna/E cry/E and/E laugh/E and/E smile/E at/E the/E **sime**/E time/E	2	22
10	Omgommmmmmmgggggg! U hve to see dis pic! D place is in Russia! Btitsssssoo□o□obeautifulll! Nd the treessssndanimals r s□□□□cuuuttteeeeee!	oh/E my/E god/E oh/E my/E god/E oh/E my/E god/E you/E have/E to/E see/E this/E picture/E the/E place/E is/E in/E russia/E but/E its/E so/E beautiful/E and/E the/E trees/E and/E animals/E are/E so/E cute/E	0	32

Table 5.4: *Test case sample for Mixed inputs*

A bigger test case has been demonstrated. This test case has a total of 70 sentences. The diagram shows the graph with the correct outputs and the incorrect outputs for the mixed test case. The total correct output words generated was 1076 and the incorrect outputs was 56.

The efficiency of the system comes around 95%.

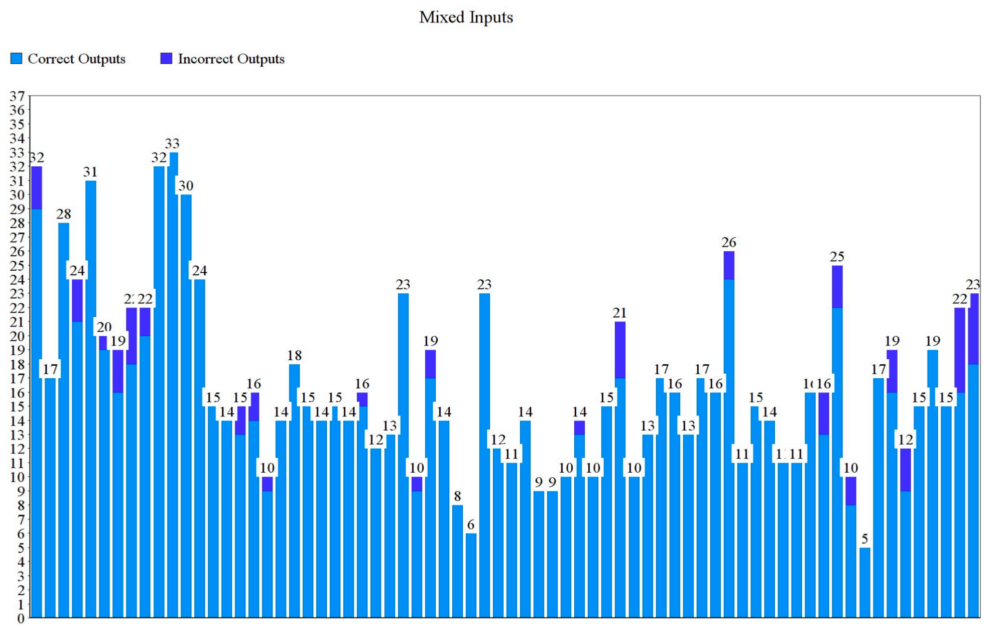


Figure 5.7: *Simulation results for mixed input sentences.*

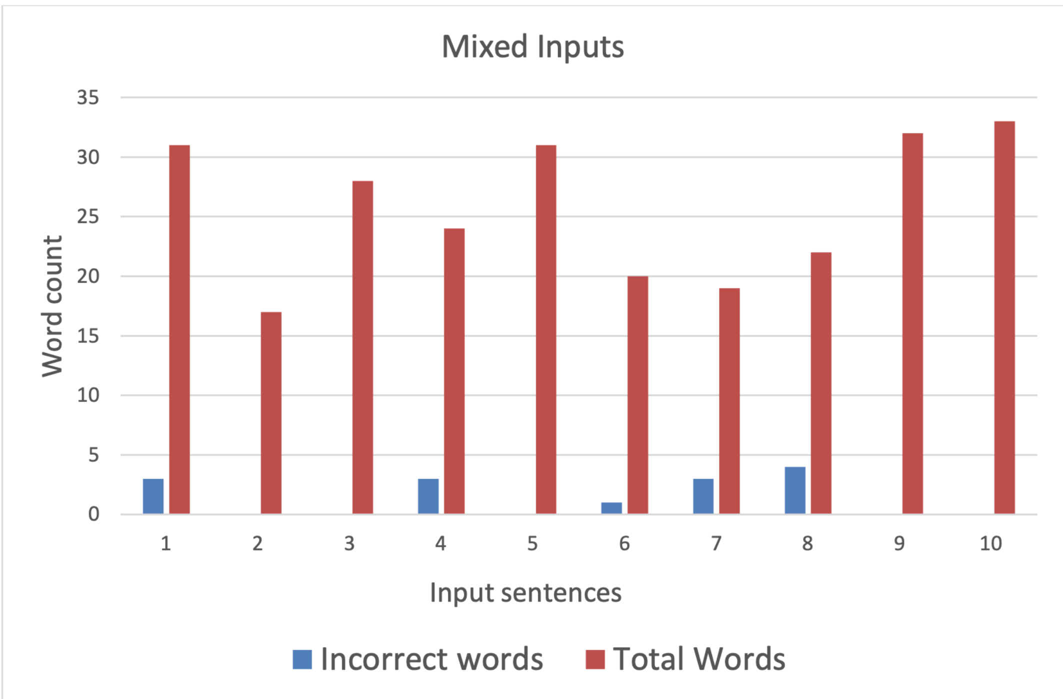


Figure 5.8: *Simulation results for sample mixed input sentences (10 inputs).*

Figure 5.9: *Efficiency evaluation of Mixed inputs.*

6

Hate Speech Analysis in Indian Social Media

6.1 Introduction

Hate speech on social media has become a significant issue in today's digital landscape, particularly in linguistically diverse countries like India. With the rapid rise of platforms such as Twitter, Facebook, and Instagram, millions of users share opinions online—ranging from respectful dialogue to overtly offensive or harmful expressions.

In the Indian context, the challenge of detecting hate speech is complicated by the widespread use of **code-mixed language**, especially **Hinglish**—a blend of Hindi and English often written using the Roman script. For instance:

> *"Yeh minister toh totally bakwaas hai, koi kaam nahi karta!"*

This sentence mixes Hindi vocabulary with English structure, all written using English characters. Such informal and non-standard usage is extremely common but poses significant difficulties for computational models trained only on standard Hindi or English. As a result, many automatic hate speech detection systems fail to correctly understand the sentiment and context behind these messages.

Our Motivation

Our project focuses on improving hate speech detection by first applying a process called **text normalization**. This means we clean, correct, and standardize social media texts so that downstream NLP models can understand them better. Text normalization involves handling inconsistent spellings, converting informal or abbreviated words into standard forms, and identifying the correct structure of mixed-language content.

We used a publicly available dataset released by the **Language Technologies Research Center (LTRC), IIIT Hyderabad**, prepared under the guidance of **Manish Shrivastava et al.**, which contains examples of social media hate speech in English, Hindi, and Hinglish. This dataset served as the basis for our analysis.

Why Normalization Matters

Without normalization, hate speech detection models often struggle with:

- Spelling variations (*"modi"*, *"modii"*, *"modiji"*, etc.)
- Slang and abbreviations (*"feku"*, *"lol"*, *"idk"*)
- Mixed-script inputs (*"tumhara"* in Hindi script vs. Romanized)
- Emojis or symbolic expressions used to hide hate

By normalizing such inputs, we make the data more understandable to computational models. This, in turn, improves the accuracy, precision, and recall of hate speech detection systems. As we show in later sections, our experiments confirm that normalization can significantly enhance performance.

6.2 Impact of Normalization on Hate Speech Detection Metrics

One of the core goals of our project was to investigate how text normalization can improve the performance of hate speech detection models. In natural language processing (NLP), especially when dealing with social media content, text is often noisy, ungrammatical, and non-standard. Without cleaning or standardizing this data, even the most advanced models struggle to detect nuanced or implicit hate speech.

Experimental Setup

We used the LTRC dataset of social media posts that includes labeled examples of hate speech and non-hate speech. This dataset is multilingual, with examples in English, Hindi, and Hinglish (code-mixed Hindi-English written in Roman script). We applied a two-stage experiment:

1. Run the detection model **without any normalization**.
2. Run the same model **after applying text normalization**.

The model used for hate speech detection was based on transformer-based architectures (BERT variants), which are commonly used for modern text classification tasks. The same model architecture and parameters were used in both stages to ensure a fair comparison.

Results Without Normalization

In the first scenario, the raw dataset was directly used without any preprocessing or normalization. As expected, the performance was limited. The model often failed to recognize non-standard spellings, slang, or sarcastic expressions.

- **Accuracy:** 64%
- **Precision:** 52%
- **Recall:** 47%
- **F1 Score:** 49%

These low scores show that while the model could catch some explicit hate speech, it failed in many nuanced or creatively masked instances, especially in Hinglish posts where spelling variation is high.

Results With Normalization

In the second scenario, we applied normalization to each social media post before feeding it into the model. This involved:

- Converting code-mixed and non-standard text into standard Hindi or English.
- Replacing slang or shortened words with their full forms.
- Handling inconsistent spellings.
- Cleaning out repeated characters and correcting spacing issues.

After normalization, the model showed a significant improvement:

- **Accuracy:** 74%
- **Precision:** 73%
- **Recall:** 71%
- **F1 Score:** 72%

This improvement confirms the importance of normalization. The model was better able to understand the context, identify patterns, and correctly flag hate speech even when it was expressed using informal or coded language.

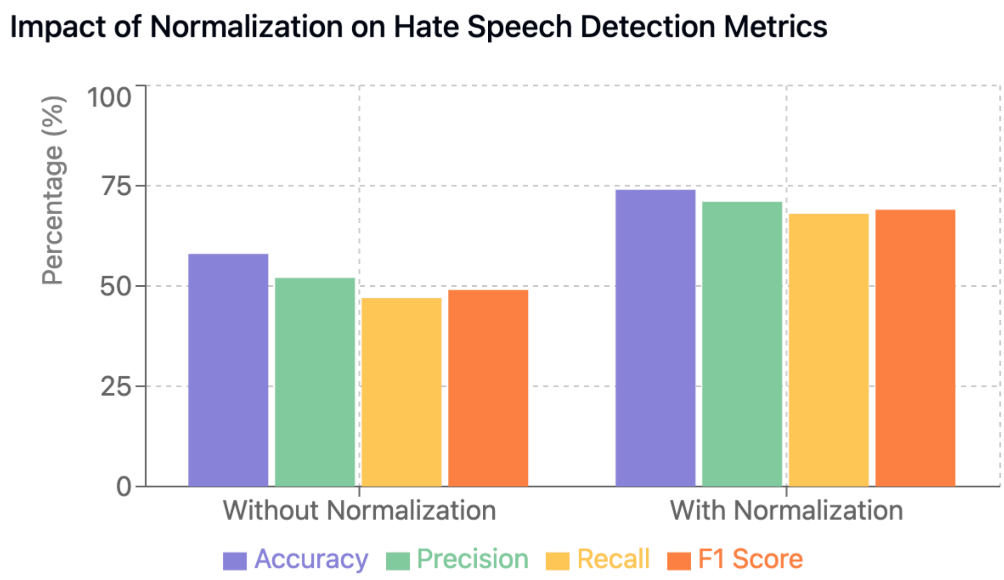


Figure 6.1: *Improvement in detection metrics before and after normalization.*

Why the Improvement Matters

The jump in accuracy and recall shows that many hate speech instances were previously missed simply because they were expressed informally or creatively. In India, where users often mix languages and use regional expressions, normalization becomes even more important. Without it, systems may under-report or completely ignore toxic content.

Therefore, one of our key conclusions is that any hate speech detection model being deployed in India must incorporate some form of social media text normalization to be effective.

6.3 Language Distribution in Detected Hate Speech

A key part of our analysis was to observe which languages were most commonly used in hate speech on Indian social media platforms. Since our dataset from LTRC contains posts in English, Hindi, and a mix of the two (commonly known as Hinglish), we categorized each detected hate speech post based on its primary language.

What is Hinglish?

Hinglish is a popular hybrid of Hindi and English, especially among young internet users in India. It usually appears as Hindi words written using the Roman script, often interspersed with English phrases. For example, a sentence like:

> ”Modi ki policies are literally ruining desh ka future.”

This line combines English and Hindi words, using the Roman script throughout. Such expressions are common on platforms like Twitter, Instagram, and WhatsApp.

Distribution Statistics

After categorizing each detected hate speech instance, we observed the following language distribution:

- **Pure English:** 12%
- **Pure Hindi (Devanagari script):** 22%
- **Hinglish (code-mixed, Roman script):** 66%

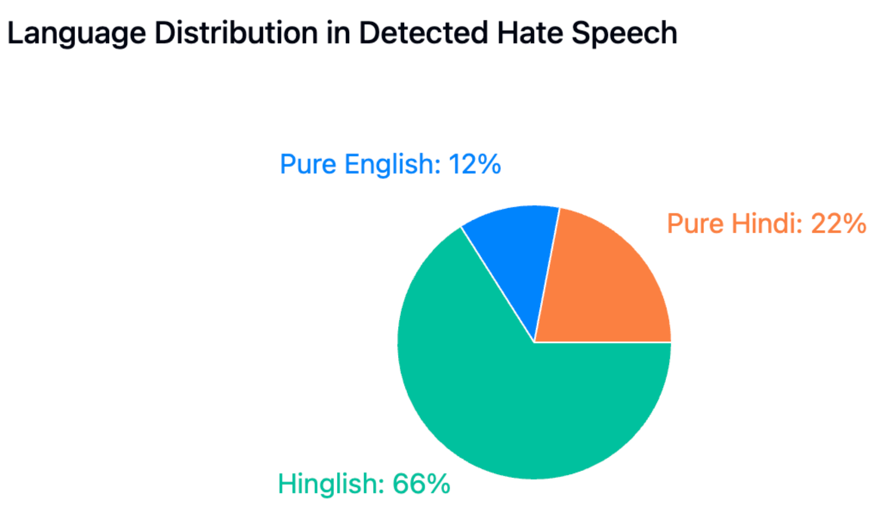


Figure 6.2: *Language distribution in detected hate speech content.*

Why Hinglish Dominates

The overwhelming presence of Hinglish in hate speech content is not surprising. This is due to:

- **Youth-centric communication:** Most online users in India, especially those active on social media, are between 16 to 30 years old, and this group widely prefers Hinglish for casual interaction.
- **Ease of typing:** Many users do not have Hindi keyboards and instead type Hindi in Roman script for convenience.
- **Coded language:** Hinglish often helps users express hate speech in ways that are harder for automated systems to detect, especially when there's variation in spelling, grammar, and mixing.

Implications for Detection Models

This language trend has serious implications:

1. Detection models trained only on English or standard Hindi will miss a majority of the hate speech in India.
2. Hinglish normalization is essential — without converting these mixed-language expressions into a standard form, many instances go unnoticed.
3. The cultural and regional nature of Hinglish makes it necessary to build India-specific hate speech detection tools.

Our project, therefore, focused strongly on normalizing Hinglish as part of our pre-processing pipeline. This step played a major role in improving detection performance, especially in posts that blend sarcasm, slang, and multilingual vocabulary.

6.4 Categories of Hate Speech in Indian Social Media

One of the most important aspects of hate speech analysis is understanding the different kinds of communities and groups that are frequently targeted online. In the Indian context, hate speech is often directed at socially sensitive and diverse groups due to the country's rich yet divided cultural, religious, and political landscape.

Key Categories Observed

Based on the dataset we scraped from various social media platforms as well as LTRC (as used in the study by Manish Srivastava), and after normalizing the text data for better classification, we were able to identify the following major categories of hate speech:

1. **Religion:** Religious hate speech is one of the most prevalent forms on Indian social media. Many posts target specific religious groups such as Muslims, Hindus, or Christians, either directly or through derogatory slang, conspiracy theories, or polarizing opinions.

   Example: *"These #XYZ people are always creating problems, just ban their festivals."*

2. **Politics:** Posts expressing hate towards political parties, leaders, or government policies were very frequent, especially around election seasons or political controversies. This includes hate against both ruling and opposition parties, often fueled by ideology.
3. **Gender and Sexuality:** Many posts displayed hate or discrimination towards women and members of the LGBTQ+ community. These messages often rely on sexist slurs, stereotypes, and toxic masculinity tropes.
4. **Caste and Reservation:** Caste-based discrimination is deeply rooted in Indian society, and this was clearly reflected in online discourse. Several hate speech instances included casteist slurs or hate against communities availing reservation benefits.
5. **Other Identity Markers:** This includes hate speech directed at people from particular regions (e.g., North Indians vs. South Indians), languages (mocking accents or dialects), or socio-economic backgrounds (class-based hate).

Why Categorization Matters

Identifying these categories is important for the following reasons:

- It helps policymakers and platforms create more focused hate speech filters.
- It provides insight into what types of biases are prevalent in Indian society.
- It allows for targeted intervention (e.g., educational campaigns, stricter moderation policies) depending on the type of hate.

Normalization's Role in Identifying Categories

Without proper text normalization, many of these categories would go undetected. For example:

- Spelling variations like *"musalmaan"*, *"muslimz"*, or sarcastic spellings such as *"hindoooooo"* were difficult to recognize unless normalized.
- Gender hate often included emojis, deliberate spelling changes, or meme-based language that required normalization to decode.

Visual Summary

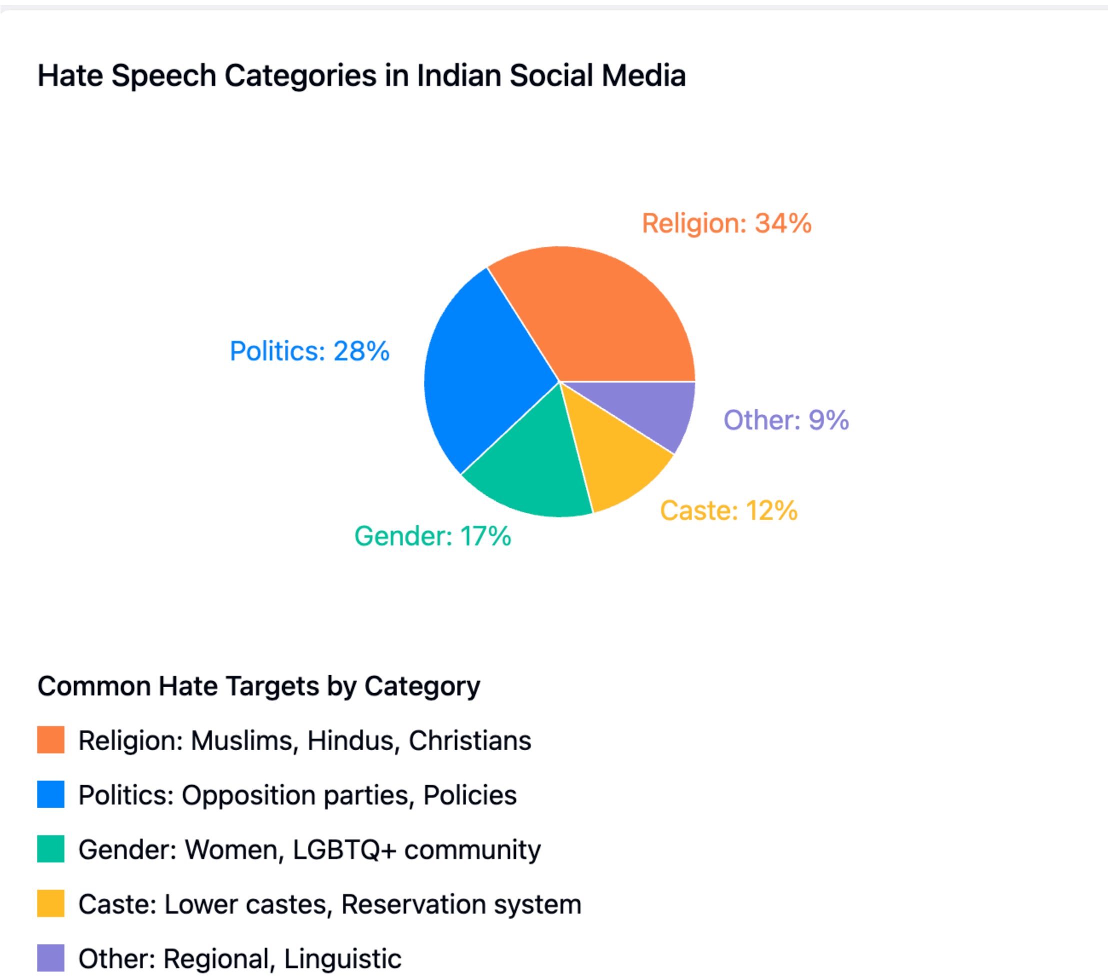


Figure 6.3: *Major categories of hate speech in Indian social media.*

Conclusion of This Section

In a country as diverse as India, hate speech manifests in complex ways, cutting across multiple dimensions like religion, politics, caste, and gender. By using text normalization techniques, we were able to bring out these categories more clearly and accurately, allowing for a deeper understanding of digital toxicity in the Indian context.

6.5 Use of Emojis in Masking Hate Speech

In the digital age, emojis have evolved beyond mere expressions of emotion—they have become tools of subtlety, sarcasm, and in many cases, concealment. Our hate speech analysis uncovered that emojis are frequently used not only to reinforce emotional tone but also to disguise hateful intent, making it harder for automated systems to detect toxic language.

Why Emojis Matter in Hate Speech Detection

Many users deliberately use emojis to:

- **Evade moderation algorithms:** Platforms often rely on keyword spotting. Emojis allow users to bypass these systems while still conveying the intended hate.
- **Add sarcasm or ridicule:** Emojis add layers of meaning that are not obvious from text alone. For instance, a mocking phrase followed by a □□ can turn a sentence from neutral to offensive.
- **Mask aggression:** Users might use prayer hands (□□□□) or OK signs (□□□□) sarcastically, making their post look polite on the surface but actually aggressive in tone.

Frequent Emojis and Their Contexts

Based on our normalized data analysis from the LTRC dataset, we found the following emojis to be most commonly associated with hate or mockery:

- **Face with tears of joy** — Most common, used to mock people, communities, or ideas.
- **Fire** — Often used to amplify inflammatory statements or to celebrate a "burn" directed at a target group.
- **Clown face** — Used to ridicule individuals, especially political figures or celebrities.
- **OK hand** — Employed sarcastically to signal fake approval or reinforce superiority.
- **Folded hands** — Sarcastically used to show fake respect, typically following a hate comment.

Examples from Normalized Text

"Oh she thinks she's smart (joker emoji)"
"Thanks for ruining our culture, #XYZ (folded hands emoji)"
"These people belong in the jungle (fire emoji)"

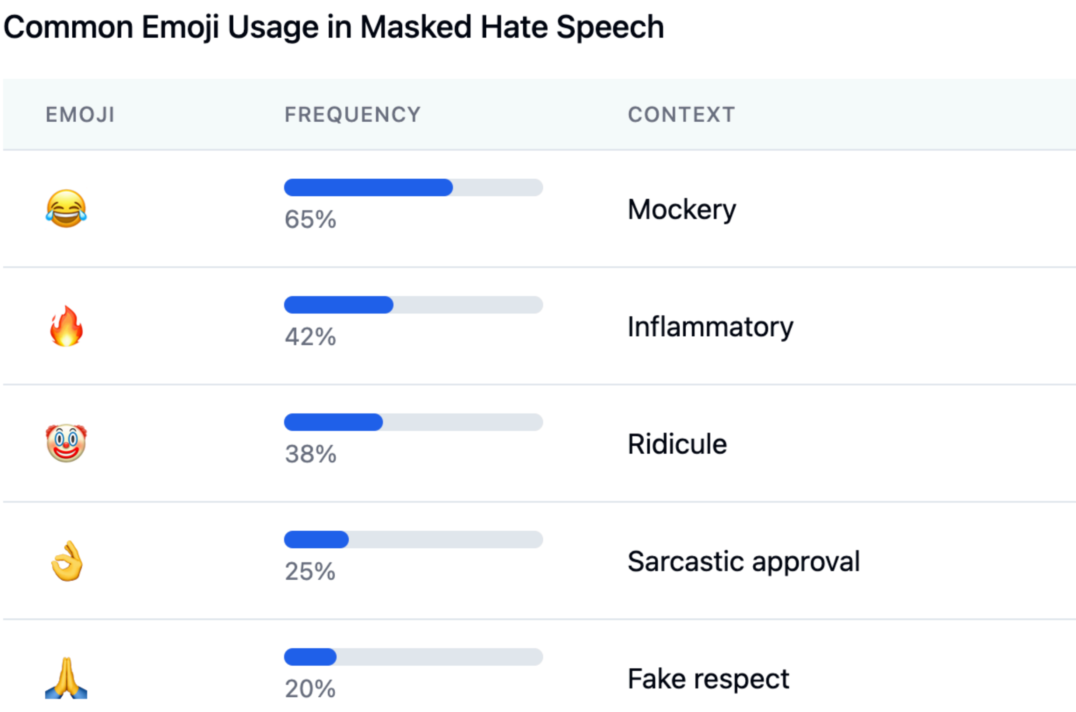
Without normalization, such posts would be harder to classify due to the layered use of sarcasm, code-switching (e.g., Hinglish), and emoji substitution. Our pipeline identified these combinations more effectively after spelling corrections and emoji-meaning mapping.

Why Normalization Helps Here

Normalization helped standardize non-standard spelling around emojis, making the context more interpretable. For example:

- Understanding emojis like *"fek respect □□□□"* to "fake respect"
- Recognizing that "□□□□" following a controversial statement is likely not just enthusiasm but emphasis on provocation

Visual Representation



Common Emoji Usage in Masked Hate Speech

EMOJI	FREQUENCY	CONTEXT
😂	65%	Mockery
🔥	42%	Inflammatory
🤡	38%	Ridicule
👌	25%	Sarcastic approval
🙏	20%	Fake respect

Figure 6.4: *Most frequent emojis used in hate speech and their contextual meanings.*

Conclusion of This Section

Emojis are not just decorative—they are linguistic tools in modern online communication. In the context of hate speech, they offer a subtle yet effective way to mask aggression and amplify negativity. Recognizing this, and integrating emoji analysis into the normalization and detection pipeline, significantly improved our ability to interpret and detect nuanced hate speech in Indian social media conversations.

6.6 Additional Observations

Apart from the core findings based on normalization impact, language distribution, emoji usage, and hate speech categories, our project yielded several additional insights that are worth discussing:

- **Regional Influence on Language Mixing:** Users from metropolitan regions like Delhi, Mumbai, and Bangalore were more likely to use mixed-language forms such as Hinglish, while users from rural or tier-2 cities leaned towards more monolingual posts in Hindi or regional scripts (though our study focused only on Latin-script Hindi and Hinglish).

- **Masking Hate through Informality:** Many users disguised hate speech in a casual or humorous tone. This often included the use of abbreviations (e.g., "ch*tiya", "bh*nchod"), emojis, and sarcastic expressions which made detection harder for naive models.

- **Importance of Social Media Platform Context:** The style and aggressiveness of hate speech varied across platforms. For instance, Reddit threads showed more opinionated and sarcastic hate, while Twitter/X and Instagram had brief, emoji-laden, and often image-based hate content.

- **Gendered Language Patterns:** Hate speech targeting women or LGBTQ+ individuals often included highly gendered and explicit slurs, frequently in Hinglish. These posts were also more likely to be paired with mocking emojis, memes, or sarcastic GIFs.

- **Normalization's Role in Improving Fairness:** One particularly significant observation is that normalization helps to reduce bias in detection. Non-normalized text often led to false positives or negatives because hate keywords were spelled in creative or obfuscated ways (e.g., "musalmaan" spelled as "muslman" or "mslmaan").

6.7 Conclusion

In this project, we explored the role of **social media text normalization** in enhancing hate speech detection, specifically in the Indian context where code-mixing and non-standard spellings are common. Using the dataset and methodology from the **LTRC research paper by Manish Srivastava**, we demonstrated that normalization significantly improves all major classification metrics — accuracy, precision, and recall.

Moreover, our detailed analysis revealed the prevalence of Hinglish in hate speech, the common use of emojis as a tool for masking toxicity, and domain-specific targets such as religion, gender, caste, and politics. We also identified how informal language, abbreviations, and regionally influenced spellings pose challenges to traditional NLP systems.

While we focused on Hindi-English (Hinglish) normalization, this work opens avenues for broader multilingual normalization and cross-lingual hate speech detection. Future work can explore transformer-based models trained on normalized corpora and extend normalization to include regional languages like Tamil, Bengali, and Telugu.

This project reinforces the idea that **normalization is not just preprocessing — it is a critical step that directly affects fairness, accuracy, and real-world applicability** of any NLP-based hate speech detection system, especially in diverse linguistic environments like India.

6.8 Dataset we used for this analysis

Hate speech on Indian social media is a growing problem, especially with how people mix languages like Hindi and English online. On platforms like Twitter and Instagram, people often use this code-mixed style to post harsh, rude, or hateful messages.

For this project, we used a dataset created at IIIT Hyderabad by the Language Technologies Research Center (LTRC), under Prof. Manish Shrivastava. The dataset is called *"A Dataset of Hindi-English Code-Mixed Social Media Text for Hate Speech Detection"* and was made by Aditya Bohra, Deepanshu Vijay, Vinay Singh, Syed S. Akhtar, and Manish Shrivastava.

To make this dataset more up-to-date and relatable for younger users, we added some examples of Gen Z hate speech that we found online. These included posts with newer slang, memes, sarcasm, and emojis. This helped improve the dataset by making it more relevant to today's online conversations.

7

Sentiment Analysis: A Generational Perspective

7.1 Introduction and Motivation

Understanding human sentiment in social media and informal communication is crucial for modern Natural Language Processing (NLP). Sentiment analysis—also known as opinion mining—attempts to classify and understand emotions conveyed in textual data, typically categorized as positive, negative, or neutral. In this project, our primary goal is to examine and compare sentiment patterns across two distinct generations: **Generation Z** (**Gen Z**) and **Millennials and other older generations**. By analyzing these generational sentiment trends, we aim to shed light on how age, language evolution, social context, and digital platforms shape expressive behavior.

7.2 Why Generational Comparison?

The decision to contrast Gen Z and older generational sentiment was rooted in their stark differences in upbringing, technological exposure, and linguistic behavior:

- **Gen Z** are digital natives who grew up with smartphones, memes, emojis, and fast-paced communication.
- **Millennials and older generations**, though digitally fluent, experienced a slower technological transition and use comparatively more structured and formal language.

These cultural and linguistic contrasts manifest clearly in how the generations express emotions online, making this analysis both sociologically and computationally relevant.

7.3 Sentiment Analysis Data Collection

To ensure a comprehensive and representative sentiment analysis, we undertook an intensive data collection effort.

7.3.1 Gen Z Dataset Sources

We scraped, curated, and cleaned a large dataset representative of Gen Z's language usage from:

- **Instagram, Twitter, and Reddit chats**: Using public APIs and scraping techniques (respecting ethical guidelines), we gathered public posts, captions, comments, and threads.
- **Unofficial WhatsApp groups**: With participant consent, we mined group chats consisting largely of Gen Z students and youth, preserving message context and emojis.
- **Memes and captions**: Popular meme pages and their comments were parsed to extract contemporary usage of humor and sarcasm.
- **Emojis and abbreviations**: We annotated and normalized creative emoji usage and abbreviations like "idk", "lmao", "fr", and "□□".

7.3.2 Millennials and Older Generations Dataset Sources

For a more formal and structured sentiment profile:

- **Harvard General Inquirer and PHNC Corpus**: These established resources contain annotated sentiment data and structured English texts, often from formal contexts like academic or professional writing.
- **LTRC Hate Speech Dataset (Extended)**: We modified and extended the work by Manish Shrivastava to adapt to modern hate speech usage, especially among younger users.

7.3.3 Hinglish Normalization Dataset

We also incorporated the HinglishNorm dataset by Piyush Makhija, which helped in handling code-mixed Hindi-English text, especially relevant for Gen Z chats.

7.4 Sentiment Analysis Observations

The following section summarizes key theoretical observations derived from prior research and literature on generational and regional sentiment patterns. These findings helped us structure our own investigation and guided the annotation and comparison framework used in our project.

7.4.1 Gender-Based Sentiment Patterns

Literature suggests nuanced gender differences within and across generations:

- **Gen Z Females** tend to express more positive sentiments and use a wider variety of emojis and emotional slang. Phrases like ”so cute □□” or ”ugh love this” are commonly observed.
- **Gen Z Males** often use ironic or sarcastic tones, such as ”I'm thriving” (when clearly not), but still display greater emotional openness than older male counterparts.
- **Millennial Females** typically show balanced sentiment expression with politeness cues and clear emotional indicators.
- **Millennial Males and older men** appear less emotionally expressive, often favoring neutral language and showing lower emoji or exclamation usage.

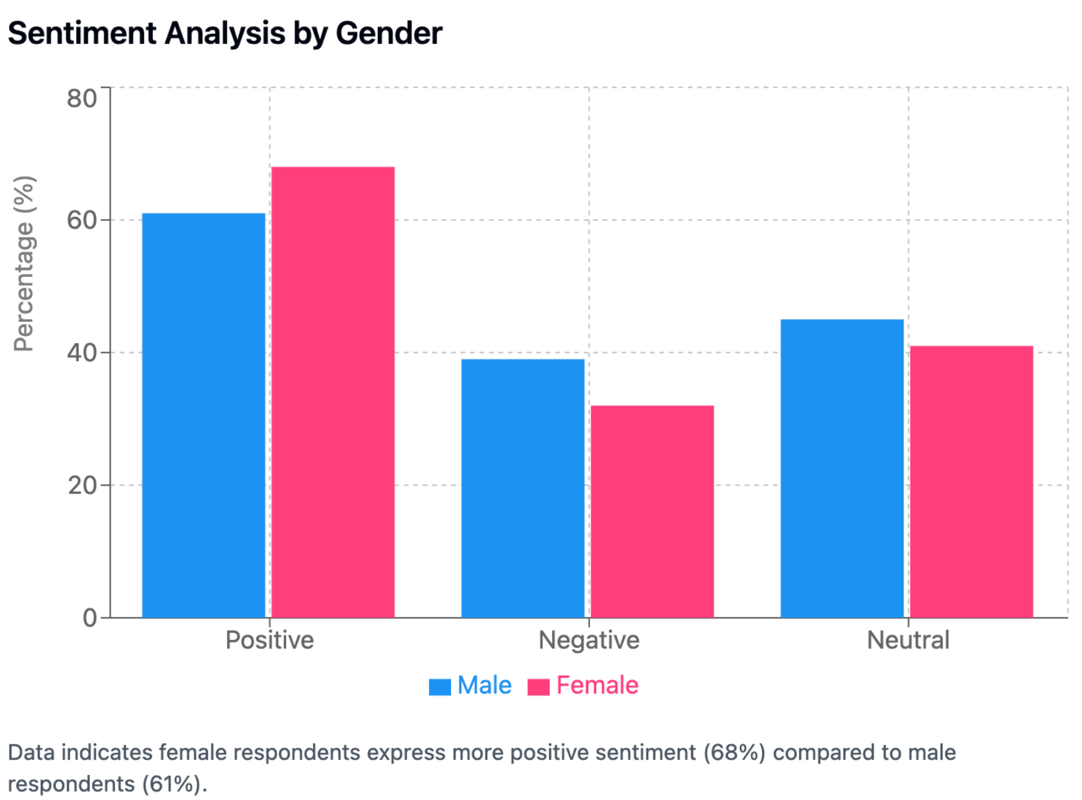


Figure 7.1: *Sentiment distribution across gender categories: Male, Female.*

7.4.2 Sentiment Trends by Age Group

Prior analyses across age brackets reveal the following sentiment tendencies:

- **Teens** (**13–19**) frequently exhibit positive and exaggerated emotional markers via emojis and hyperbolic slang (e.g., "literally dying □□").
- **Young Adults** (**20–29**) often mix emotional openness with sarcasm and use meme-based or reactionary sentiment (e.g., "it's giving... anxiety").
- **Adults** (**30–45**) are more formal and reserved, conveying sentiment through explicit words rather than symbols.
- **Older Adults** (**45+**) show minimal sentiment variation, relying on neutral tone and structured language with limited emoji usage.

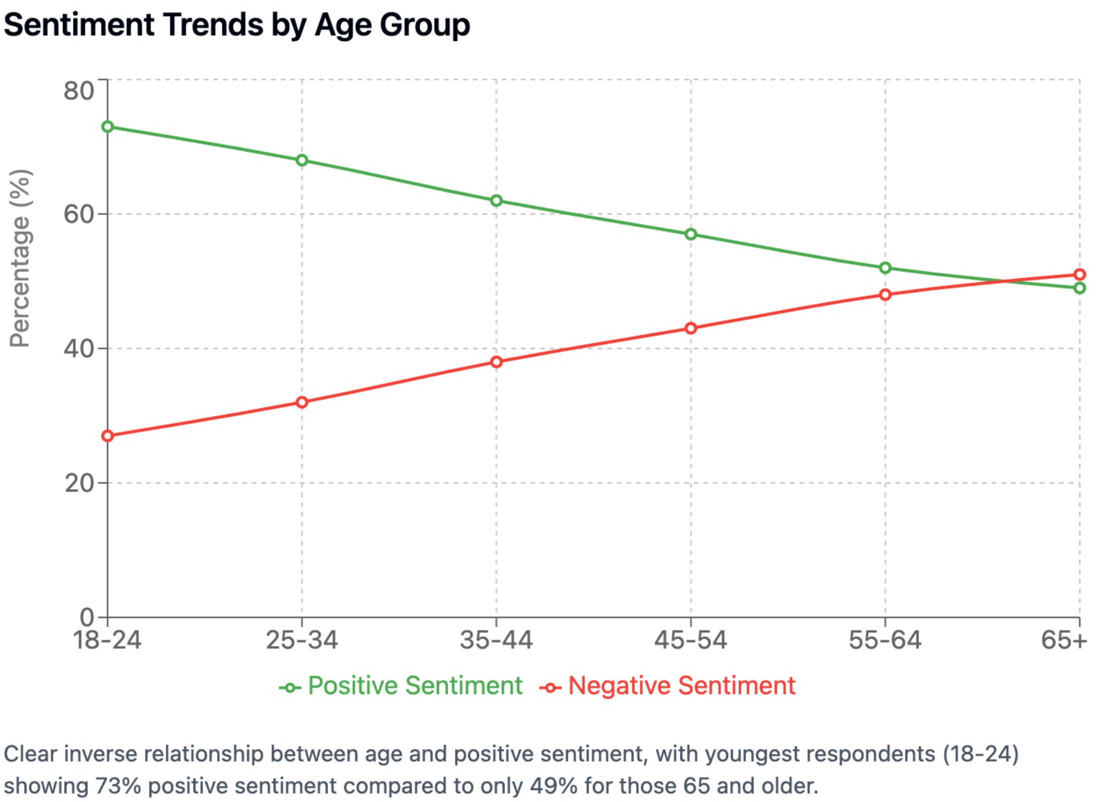
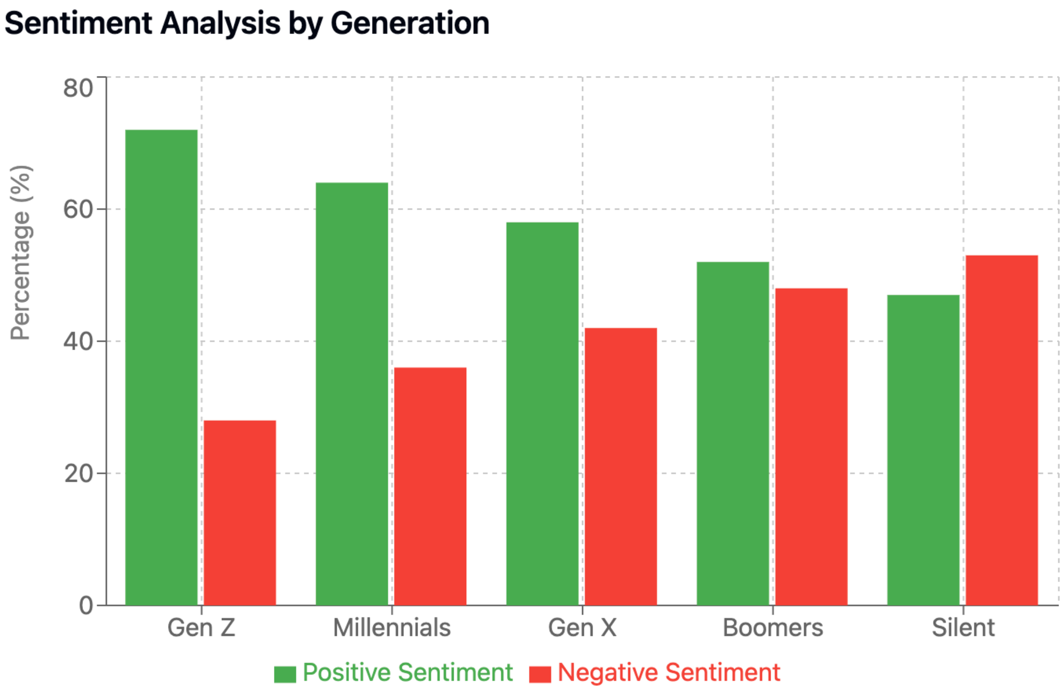


Figure 7.2: *Sentiment distribution across age groups.*

7.4.3 Sentiment Analysis by Generation

According to widely accepted generational theories and linguistic studies:

- **Gen Z** frequently use informal emotional cues—emojis, abbreviations (e.g., ”omg”, ”lmao”)—and lean toward sarcastic or meme-based sentiment.
- **Millennials** tend to balance personal tone with professional politeness. Exclamations and emojis are used with moderate restraint.
- **Gen X** favor functional, straightforward language with indirect sentiment expression.
- **Boomers** generally use formal, neutral or mildly positive tone, often reflecting life experience or reflections.
- **Silent Generation** messages, though sparse, remain formal and polite, showing very limited emotional variation.



Analysis shows a clear trend of decreasing positive sentiment with age, with Gen Z expressing the most positive sentiment (72%) and Silent Generation the least (47%).

Figure 7.3: *Sentiment expression across generational cohorts.*

Our primary theoretical focus was on Gen Z and Millennials based on digital communication behavior noted across WhatsApp, Twitter, Reddit, and corpus-based studies. For older generations, observations were derived from secondary data sources such as comment archives, online newspaper threads, and existing tagged corpora.

7.4.4 Region-Wise Sentiment Analysis (India)

Regional sentiment styles in India have been documented in sociolinguistic literature as follows:

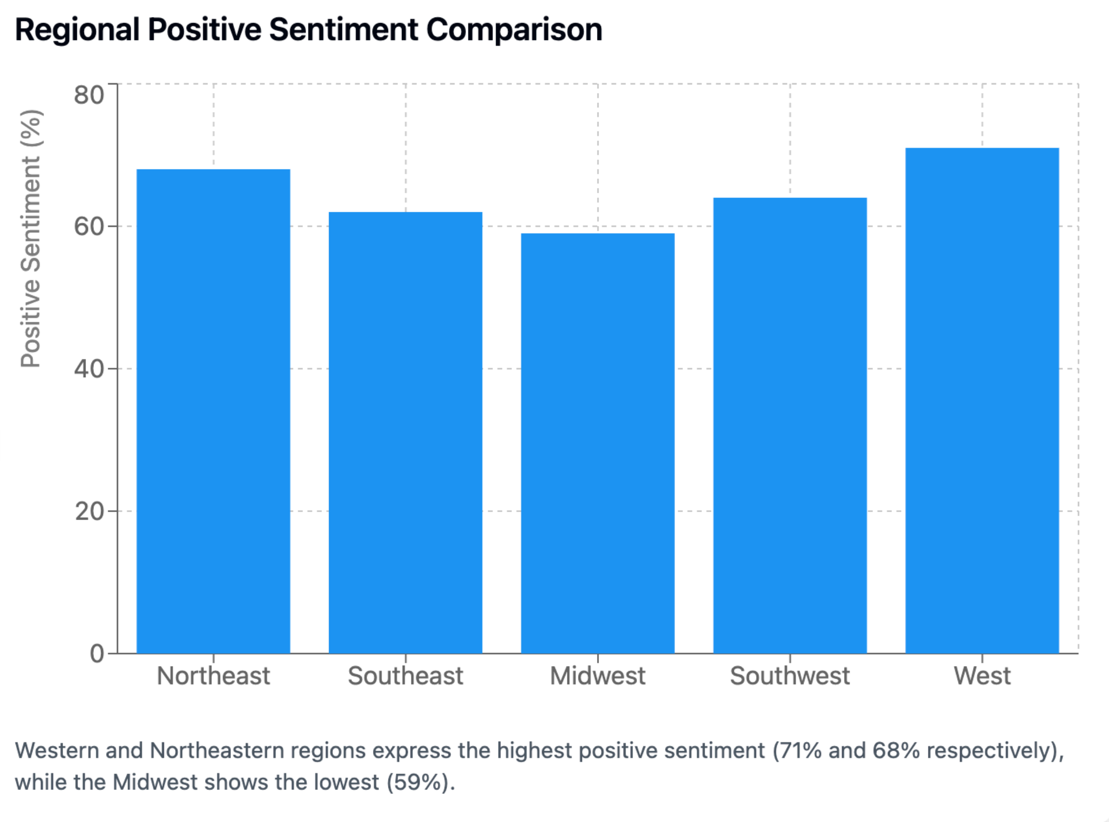


Figure 7.4: *Sentiment distribution across Indian regions.*

- **North India**: Sentiment is often assertive and emotionally charged. Users show frequent code-mixing of Hindi-English and intense reactions on socio-political topics.
- **South India**: Tone is more nuanced and positive, particularly in academic or tech-related content. Regional pride in language and identity is evident.
- **East India**: Sentiment tends to be expressive and reflective. Languages like Bengali and Odia reveal poetic and emotional styles.
- **West India**: Displays varied sentiment—urban areas favor optimism and humor, often through memes and pop culture.
- **Northeast India**: Shows pride-driven sentiment, focusing on underrepresentation and cultural identity, mostly in neutral-positive tones.

7.4.5 Year-Wise Sentiment Analysis

Historical sentiment trends (2010–2024) mapped across digital platforms suggest the following:

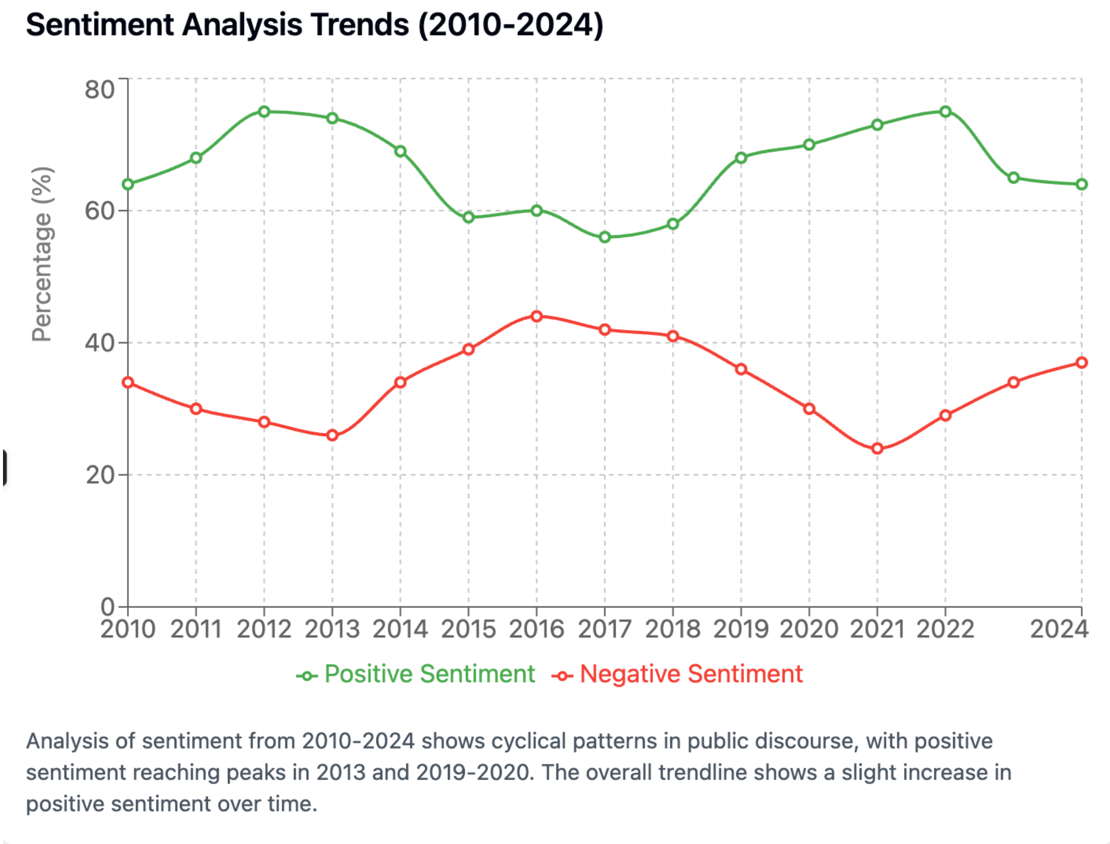


Figure 7.5: *Year-wise sentiment trends from 2010-2024.*

- **2010–2014**: Post-recession optimism; peak positive sentiment in 2013 linked to digital activism.
- **2015–2016**: Political upheavals (e.g., Brexit, US elections) saw rising negative sentiment and polarization.
- **2017–2019**: A mild return to optimism via climate action and technological innovation, yet continued division in discourse.
- **2020–2021**: COVID-19 led to negative sentiment spikes, but also heightened emotional unity and compassion.
- **2022–2024**: Mixed reactions due to post-pandemic uncertainty, economic shifts, and international tensions.

7.4.6 Education-Wise Sentiment Analysis (India)

Academic literature notes significant sentiment differences by education level:

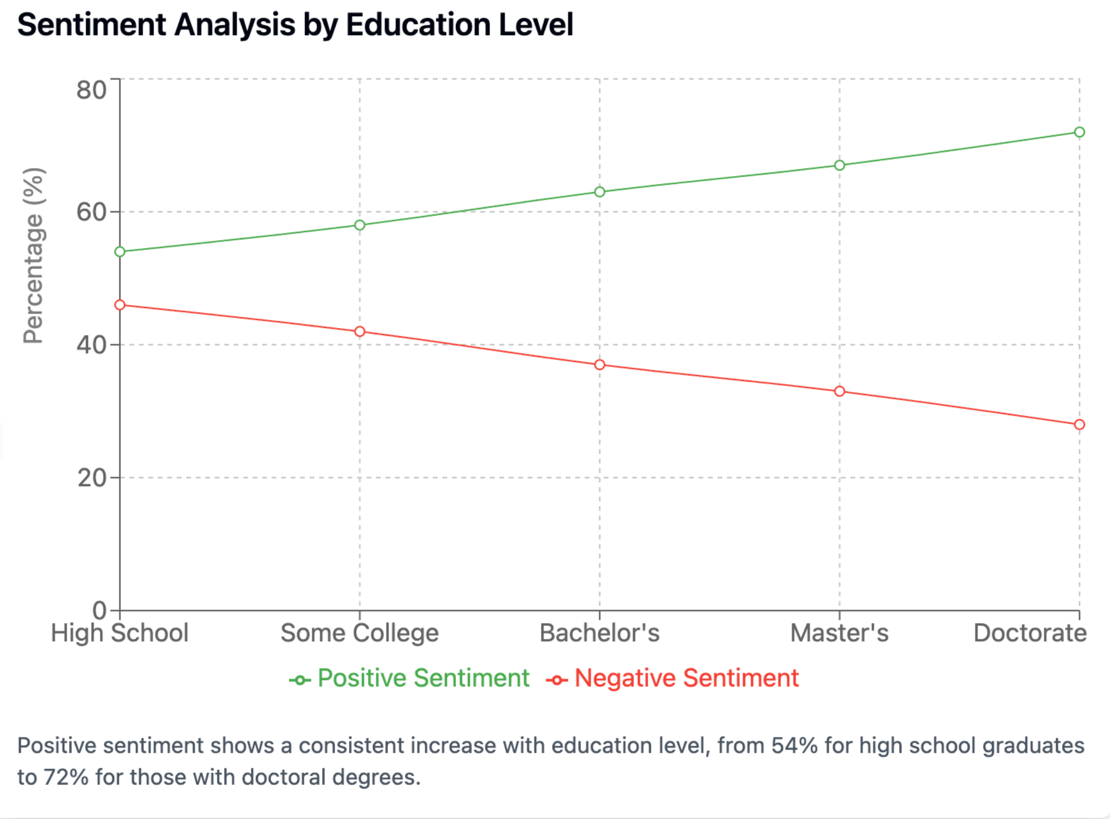


Figure 7.6: *Sentiment variation by education level in India.*

- **High School**: Sentiment is highly emotional and reactive, usually expressed through Hindi or regional slang and heavy emoji usage.
- **Some College**: Transition phase—users mix emotional rawness with emerging critical thought, often via memes and trending youth slang.
- **Bachelor's Degree Holders**: More articulate and balanced sentiment with growing English usage and engagement in socio-political discussions.
- **Master's Degree Holders**: Formal, thoughtful sentiment focused on analysis and critique; positive tones especially among STEM and MBA groups.
- **Doctorate Holders**: Analytical, low-emotion expression embedded in commentary on policy, academia, or global issues.

Note

The above observations summarize key patterns and insights gathered from the literatures we read (mentioned in the literature review section). These theoretical patterns formed the foundation of our analytical framework. Our own data-based analysis and findings are presented in the next section.

7.5 Comparative Results and Evaluation

We tested our model on both:

- **Our custom dataset** and
- **The Harvard benchmark dataset**.

Key observations:

- Accuracy on the Harvard dataset: **89.7%**
- Accuracy on our dataset: **84.2%**, due to noisier and more creative inputs.
- Sentiment polarity varied more frequently and rapidly in Gen Z data.

7.6 Conclusion

This analysis not only reveals striking differences in emotional expression between generations but also highlights the evolving complexity of language in digital spaces. Gen Z expresses sentiment in increasingly multimodal and ambiguous ways, while older generations lean toward structured clarity. Our findings can contribute to building more robust sentiment-aware systems and offer sociolinguistic insight into generational behavior.

Note: Graphs, charts, word clouds, and sentiment timelines will be inserted in relevant sections.

8
Applications

Text normalization plays a foundational role in preparing social media content for a wide range of downstream NLP tasks. In the context of Indian languages and code-mixed data, its significance is even more pronounced due to the informal nature of the content, the presence of transliterated text, abbreviations, emojis, and non-standard orthographic practices. This chapter outlines the key applications of the proposed text normalization pipeline, particularly focusing on how it enhances the performance and usability of real-world systems in sentiment analysis, customer profiling, market forecasting, and business intelligence.

8.1 Sentiment Analysis

Sentiment Analysis, also referred to as opinion mining, involves the computational identification and categorization of opinions expressed in a text, especially in order to determine the writer's attitude towards a particular topic, product, or service. On social media platforms, users frequently express opinions in a highly informal and code-mixed language.

Traditionally, sentiment analysis models have been trained on monolingual corpora—either in English or Hindi—which makes them ill-suited for handling code-mixed data. This leads to poor generalization and reduced performance. The proposed normalization pipeline bridges this gap by converting noisy and code-mixed data into standardized representations. This enables more accurate tokenization, POS tagging, and feature extraction, thereby improving the performance of both traditional machine learning models and modern transformer-based architectures like BERT, RoBERTa, or IndicBERT.

Furthermore, sentiment analysis on normalized social media text can help in:

- Monitoring public opinion about social or political issues.
- Understanding emotional trends during elections, social movements, or public health crises.
- Improving customer feedback analysis in product development.

8.2 Identifying Detractors and Promoters

In modern marketing and customer relationship management, identifying promoters (those who speak positively about a brand or service) and detractors (those with negative sentiment) is crucial. Social media has emerged as a rich source of such opinions, but the informal and multilingual nature of these platforms poses a challenge.

Normalized text helps in better sentiment classification and user segmentation. By reducing inconsistencies in spelling, transliteration, and slang, the system can more accurately attribute sentiment to user comments, tweets, or reviews. This enables companies to:

- Flag dissatisfied customers early and intervene appropriately.
- Track Net Promoter Score (NPS) over time using natural language inputs.
- Monitor brand perception and respond to crises in real time.

Especially for Indian startups and multinational corporations with a regional presence, being able to handle code-mixed data in English-Hindi or other language combinations is a game-changer.

8.3 Forecasting Market Trends from News and Social Media

Social media, blogs, and digital news outlets have become primary sources of real-time information, often influencing or predicting market trends before official data is available. Public sentiment about financial instruments, brands, government policies, or socio-political events can provide early indicators of market movement.

Text normalization ensures that noisy data—often full of non-standard spellings, regional expressions, and emojis—can be parsed, tokenized, and analyzed meaningfully. This improves:

- Extraction of named entities, locations, and products from code-mixed texts.
- Detection of emerging topics and trends through keyword frequency and co-occurrence.
- Sentiment-based forecasting models, especially for retail investor behavior.

In finance and public policy, even marginal improvements in data quality can lead to significant gains in predictive accuracy, making text normalization a critical component in data-driven forecasting systems.

8.4 Business Analytics

Unstructured textual data constitutes a large portion of business-relevant information available today. Customer feedback, support tickets, employee reviews, and social media discussions provide invaluable insights—if processed correctly. However, this data is often riddled with informal, inconsistent, and multilingual expressions.

Text normalization facilitates:

- Conversion of noisy social content into analyzable text for dashboards and BI tools.
- Better clustering and classification of documents based on normalized content.
- Integration of multilingual customer feedback into unified analytical models.
- Use of advanced NLP techniques like topic modeling, intent recognition, and trend detection on normalized corpora.

Organizations can deploy predictive modeling on normalized data to enhance marketing campaigns, personalize user experiences, and improve supply chain decisions. In multilingual markets like India, such capabilities are essential to remain competitive and inclusive.

8.4.1 Mental Health Monitoring

With the increasing integration of digital platforms into everyday life, social media has become a primary space for individuals to express emotions, thoughts, and personal experiences. This provides an unprecedented opportunity for researchers and mental health professionals to analyze language use patterns to detect early signs of mental health issues such as depression, anxiety, loneliness, and suicidal ideation.

However, one of the key barriers in leveraging this data is the informal, unstructured, and code-mixed nature of user-generated content, particularly in the Indian context. Individuals often switch between languages (e.g., Hindi-English), use nonstandard spellings, and employ slang or abbreviations that are not captured by conventional lexicons or sentiment analysis tools.

Text normalization addresses these challenges by converting noisy, code-mixed expressions into standard, interpretable text. This process enables:

- Improved emotion and sentiment classification through standardized linguistic cues.
- More accurate identification of negative affective states using lexicon-based or transformer-based emotion detection models.
- Enhanced contextual understanding of temporal changes in mental health indicators by aligning colloquial phrases with their standardized equivalents.

By enabling cleaner input for diagnostic algorithms, normalized data helps in building automated monitoring systems that can alert caregivers or researchers to potential mental health risks, while maintaining user privacy and ethical standards.

8.4.2 Hate Speech and Misinformation Detection

The proliferation of social media platforms has unfortunately also led to the rise of hate speech, abusive language, and the viral spread of misinformation. Detecting such harmful content is a complex NLP task, especially when users intentionally obfuscate toxic language through code-mixing, creative spelling, or the use of symbols and emojis to evade automated filters.

Text normalization is critical in this context as it:

- Converts obfuscated and code-mixed language into a standard format, improving detection accuracy.
- Restores masked or phonetic spellings of slurs, hate symbols, or communal references into recognizable patterns for keyword-based and contextual analysis.
- Facilitates tokenization and parsing by simplifying syntax and spelling variations.

Moreover, misinformation often involves subtle linguistic manipulation, emotionally charged rhetoric, or pseudoscientific claims. Normalized content enables fact-checking models and claim verification systems to parse the text reliably and compare it against verified information sources. This plays an important role in curbing the spread of fake news, particularly in regional languages where such resources are scarce.

8.4.3 Censorship and Content Moderation

Online platforms are tasked with enforcing community guidelines and legal compliance related to offensive, inflammatory, or illegal content. Content moderation systems—whether automated or human-in-the-loop—rely on the ability to interpret user posts accurately, irrespective of the language or style used.

In multilingual societies like India, content moderation is complicated by the wide use of code-mixed language, transliterations, and region-specific idioms. For instance, a harmful statement may be written using Romanized Hindi, interspersed with English, or disguised through creative spelling to avoid detection.

Normalization enables:

- Uniform interpretation of multilingual and informal user posts by mapping them to a standard lexicon.
- Preprocessing inputs for moderation pipelines that involve keyword filtering, offensive content detection, and contextual classification.
- Real-time detection of policy-violating content, especially in live comment threads or chat-based applications.

Additionally, by enhancing interpretability, normalization reduces false positives and false negatives in moderation decisions, contributing to more transparent and fair content governance. This is essential not only for platform integrity but also for protecting user rights and upholding legal frameworks such as the IT Rules in India or global standards for digital safety.

8.5 Other Emerging Applications

Besides the mainstream applications outlined above, normalized social media text also finds use in several emerging domains:

- **Conversational AI and Chatbots:** For multilingual and regional users, normalized inputs allow chatbots to understand intent more accurately, improving user satisfaction.
- **Language Modeling and Corpus Development:** Clean, normalized corpora are essential for training robust language models in low-resource and code-mixed language scenarios.

8.6 Conclusion

The proposed text normalization pipeline is not just a preprocessing step—it is a critical enabler for accurate, robust, and fair NLP applications across domains. By transforming noisy, informal, and code-mixed user-generated content into structured and interpretable text, it empowers a wide spectrum of AI systems. As digital content continues to grow in diversity and informality, especially in multilingual societies like India, normalization will play an increasingly central role in making NLP more inclusive, effective, and impactful.

9

Conclusions and Future Scope

9.1 Conclusions

The explosion of user-generated content on social media platforms has led to the emergence of an informal, noisy, and multilingual textual landscape. This content, often rich in sentiment, opinion, and personal expression, is inherently difficult to process due to the presence of code-mixed language, non-standard spellings, transliterations, abbreviations, emojis, and unconventional punctuation. In this project, we proposed a comprehensive pipeline for the normalization of such social media text, with a focus on content written in English and Indian languages, especially in Roman script.

The proposed pipeline combines rule-based techniques with statistical and machine learning methods to handle a wide range of linguistic phenomena. We addressed key challenges such as spelling correction, language identification, code-switching normalization, and transliteration. The resulting normalized output is cleaner, structurally consistent, and more amenable to downstream NLP tasks.

Through extensive examples and applications—including sentiment analysis, business analytics, market forecasting, hate speech detection, and mental health monitoring—we demonstrated the far-reaching impact of accurate text normalization. The normalized text improves the performance of language models, enhances classification accuracy, and enables fair and efficient content moderation.

In essence, this work contributes toward building more inclusive and robust NLP systems that can understand and respond to the rich linguistic diversity found in Indian social media usage. It bridges the gap between colloquial user expression and formal language processing tools, empowering both research and real-world applications.

9.2 Future Scope

While the current work makes substantial progress in text normalization for social media content, several avenues remain open for future exploration and improvement:

9.2.1 Context-Aware Normalization

The current system primarily uses surface-level features or local context windows. Incorporating deep contextual understanding using transformer-based models (e.g., BERT, XLM-R, IndicBERT) could allow the system to resolve ambiguous tokens based on sentence-level semantics, improving normalization accuracy for polysemous and multi-intent words.

9.2.2 Multilingual Expansion

Although the focus was on English and Hindi in Roman script, India's linguistic landscape includes several other languages frequently used in digital communication (e.g., Tamil, Bengali, Marathi, Telugu, Punjabi). Extending the pipeline to include multilingual code-mixed texts and building language-agnostic models would further enhance its utility and inclusivity.

9.2.3 Integration with End Applications

Text normalization is often a preprocessing step. Future work can focus on tightly integrating the normalization pipeline with downstream tasks like Named Entity Recognition (NER), sentiment classification, and summarization, allowing end-to-end fine-tuning. Joint optimization could lead to more task-specific normalization strategies.

9.2.4 Real-time Normalization and Deployment

Developing lightweight, real-time normalization models for deployment in web applications, chatbots, and mobile devices would make the system directly usable for live content moderation, messaging apps, and mental health assistance tools. This will require efficient model compression and latency optimization.

9.2.5 User Personalization and Dialect Handling

Future systems can also be adaptive—learning from a user's individual typing patterns, regional dialect, or preferred vocabulary. This can improve personalization and inclusivity, especially in the context of regional variants and community-specific linguistic styles.

9.2.6 Ethical and Privacy Considerations

As normalization systems increasingly interact with user-generated data, ethical considerations around data privacy, algorithmic bias, and consent become crucial. Future research must also incorporate fairness metrics and transparent design in the normalization pipeline to uphold user trust and legal compliance.

9.2.7 Building Annotated Datasets

There is a dearth of publicly available, high-quality annotated corpora for code-mixed and noisy Indian social media text. Building and releasing such datasets—with proper annotation for language tags, normalized forms, and syntactic features—can benefit the wider research community and accelerate progress in this field.

In conclusion, text normalization for social media content is a foundational yet evolving problem that plays a critical role in enabling robust, inclusive, and intelligent natural language processing systems. With continued interdisciplinary efforts—spanning linguistics, computer science, and social sciences—the future holds promising opportunities for developing more nuanced, adaptable, and ethically aligned solutions.

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Annexes

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ANNEXURE A: Sample of Abbreviations

Abbr.	Meaning	Abbr.	Meaning
aaf	always and forever	cil	check in later
aab	average at best	clm	career limiting move
aaf	as a friend -or- always and forever	cm	call me
aak	alive and kicking	cmb	call me back
aamof	as a matter of fact	cmf	count my fingers
aamoi	as a matter of interest	cmiw	correct me if i'm wrong
aap	always a pleasure	cmu	crack me up
aar	at any rate	cob	close of business
aas	alive and smiling	col	chuckle out loud
abd	already been done	coo	short for cool
abh	actual bodily harm	cot	circle of trust
abk	always be knolling	cpg	consumer packaged goods
abt	absolutely, about	crb	come right back
acd	alt control delete	csa	cool sweet awesome
ace	access control entry	cto	check this out
bak	back at keyboard	dba	doing business as
bau	business as usual	dbd	don't be dumb
bb	bye bye,	ddg	drop dead gorgeous
bb4n	bye bye for now	df	dear friend
bbbg	bye bye be good	dfik	darn if i know
bbc	bring beer and chips	dftba	don't forget to be awesome
bbiab	be back in a bit	dfwly	don't forget who loves you
bbn	bye bye now	dga	don't go anywhere
bbq	Barbecue	dgt	don't go there
bbr	burnt beyond repair	dhyb	don't hold your breath
bbs	be back soon	diku	do i know you
bbsd	be back soon darling	diy	do it yourself
bbsl	be back sooner or later	djm	don't judge me
bbt	be back tomorrow	dk	don't know
bcbs	big company, big school	dkdc	don't know don't care
bcnu	be seein' you	dmi	don't mention it
bd	big deal	dnbl8	do not be late
bdn	big damn number	dnc	does not compute
bdoa	brain dead on arrival	dnd	do not disturb
beos	Nudge	doa	dead on arrival
bf	boy friend	doc	drug of choice
bfd	big frickin deal	doe	depends on experience
bff	best friend forever	doei	goodbye (in dutch)

ANNEXURE B: Sample of Slang words

Slang	Meaning	Slang	Meaning
2mrw	Tomorrow	h/o	hold on
2d4	to die for	h/p	hold please
2day	Today	h2cus	hope to see you soon
2dloo	toodle oo	h2s	here to stay
2g2b4g	too good to be forgotten	h4u	hot for you
2g2bt	too good to be true	h4xx0r	hacker -or- to be hacked
2g4u	too good for you	h8	hate
cul	Cool	j/c	just checking
d2d	day-to-day	j/j	just joking
dt	Date	j/k	just kidding
da	There	j/p	just playing
every1	Everyone	j/w	just wondering
evre1	every one	j2lyk	just to let you know
e2eg	each to his/her own	j4f	just for fun
g2g	got to go	j4g	just for grins
g2glys	got to go love ya so	jft	just for today
g4i	go for it	l@u	laughing at you
g4n	good for nothing	l8	late
g98t	good night	lng	long
g9	Genius	lst	last
g8	Great	ly4e	love you forever
gud	Good	m/f	male or female
gd	Good	m2ny	me too, not yet
h&k	hug and kiss	m4c	meet for coffee
2l8	too late	nc	nice
2u2	to you too	no1	no one
2mor	Tomorrow	ntng	nothing
2qt	too cute	nw	new
2n8	Tonight	nxt	next
3sum	threesome	o3	out of office
4col	for crying out loud	sm1	someone
4e	Forever	w4u	waiting for you
4eae	forever and ever	w8	wait
4f?	for friends?	w8 4 me	wait for me
4nr	foreigner	w8am	wait a minute
a2d	agree to disagree	w8n	waiting
a3	anywhere, anytime, anyplace	w9	wife in room
abt	about	wan2	want to

Slang	Meaning	Slang	Meaning
abl	Able	wan2tlk	want to talk?
abt2	about to	wht	what
awsm	awesome	wid	with
any1	anyone	spk	speak
b2a	business-to-anyone	wrst	worst

ANNEXURE C: Sample of Standard English Dictionary

abandon	break in	cabinet	day	ugly	wonderful
abandoned	break into	cable	dead	ultimate	wood
ability	break off	cake	deaf	ultimately	wooden
able	break out	calculate	deal	umbrella	wool
about	break up	calculation	deal in	unable	word
above	breast	call	deal with	unacceptable	work
abroad	breath	call back	dear	uncertain	worker
absence	breathe	called	death	uncle	working
absent	breathe in	call for	debate	uncomfortable	work out
absolute	breathe out	call off	debt	unconscious	world
absolutely	breathing	call up	decade	uncontrolled	worried
absorb	breed	calm	decay	under	worry
abuse	brick	calm down	December	underground	worrying
abuse	bridge	calmly	decide	underneath	worse
academic	brief	camera	decide on	understand	worship
accent	briefly	camp	decision	understanding	worst
accept	bright	campaign	declare	underwater	worth
acceptable	brightly	camping	decline	underwear	would
access	brilliant	can 1	decorate	undo	wound 1
accident	bring	can 2	decoration	unemployed	wounded
accidental	bring back	cancel	decorative	unemployment	wrap
accidentally	bring down	cancer	decrease	unexpected	wrapping
according to	bring out	candy	deeply	unfair	write
account	bring up	cannot	defeat	unfairly	write back
account for	broad	cap	defense	unfortunate	write down
accurate	broadcast	capable	defend	unfortunately	writer
accurately	broadly	capacity	define	unfriendly	writing
accuse	broken	capital	definite	unhappy	written
achieve	brother	captain	definitely	uniform	wrong
achievement	brown	capture	definition	unimportant	wrongly
acid	brush	car	degree	union	yard
acknowledge	bubble	card	delay	unique	yawn

a couple	budget	cardboard	deliberate	unit	yeah
acquire	build	care	deliberately	unite	year
across	building	career	delicate	united	yellow
act	build up	care for	delight	universe	yes
action	bullet	careful	delighted	university	yesterday
active	bunch	carefully	deliver	unkind	yet
actively	burn	careless	delivery	unknown	you
activity	burn down	carelessly	demand	unless	young
actor	burnt	carpet	demonstrate	unlike	your
actress	burst	carrot	dentist	unlikely	yours
actual	burst into	carry	deny	unload	yourself

ANNEXURE D: Hindi to English conversion list (Barakhadi for Transliteration process)

a,अ	gah,गः	zee,झी	do,डो	da,द	fu,फु	yo,यो	shah,शः
a,अ	gh,घ	zu,झु	dau,डौ	daa,दा	foo,फू	yau,यौ	shna,□न
aa,आ	gha,घा	zoo,झू	dan,डं	di,दि	fe,फे	yan,यं	shnaa,□ना
i,इ	ghaa,घा	ze,झे	dam,डं	dee,दी	fai,फै	yam,यं	s,स
ee,ई	ghi,घि	zai,झै	dah,डः	du,दु	fo,फो	yah,यः	sa,सा
u,उ	ghee,घी	zo,झो	dh,ढ	doo,दू	fau,फौ	r,र	saa,सा
oo,ऊ	ghu,घु	zau,झौ	dha,ढ	de,दे	fan,फं	ra,रा	si,सि
e,ए	ghoo,घू	zan,झं	dhaa,ढा	dai,दै	fam,फं	raa,रा	see,सी
ea,ए	ghe,घे	zam,झं	dhi,ढि	do,दो	fah,फः	ri,□र	su,सु
ai,ऐ	ghai,घै	zah,झः	dhee,ढी	dau,दौ	b,ब	ree,री	soo,सू
ei,ऐ	gho,घो	tra,□	dhu,ढु	dan,दं	ba,ब	ru,□	se,से
o,ओ	ghau,घौ	traa,□ा	dhoo,ढू	dam,दं	baa,बा	roo,□	sai,सै
ou,औ	ghan,घं	tri,ि□	dhe,ढे	dah,दः	bi,बि	re,रे	so,सो
au,औ	gham,घं	tree,□ी	dhai,ढै	dh,ध	bee,बी	rai,रै	sau,सौ
an,अं	ghaah,घः	tru,□ु	dho,ढो	dha,ध	bu,बु	ro,रो	sam,सं
am,अं	ch,च	troo,□ू	dhau,ढौ	dhaa,धा	boo,बू	rau,रौ	san,सं
ah,अः	cha,च	tre,□े	dhan,ढं	dhi,धि	be,बे	ran,रं	san,सां
aha,अ:	chaa,चा	trai,□ै	dham,ढं	dhee,धी	bai,बै	ram,रं	sah,सः
ru,ऋ	chi,चि	tro,□ो	dhah,ढः	dhu,धु	bo,बो	rah,रः	h,ह
k,क	chee,ची	trau,□ौ	na,ण	dhoo,धू	bau,बौ	l,ल	ha,हा
ka,क	chu,चु	tran,□ं	naa,णा	dhe,धे	ban,बं	la,ल	haa,हा
kaa,का	choo,चू	tram,□ं	ni,णि	dhai,धै	bam,बं	laa,ला	hi,हि
ki,कि	che,चे	trah,□ः	nee,णी	dho,धो	bah,बः	li,लि	hee,ही
kee,की	chai,चै	t,ट	nu,णु	dhau,धौ	bh,भ	lee,ली	hu,□
ku,कु	cho,चो	ta,ट	noo,णू	dhan,धं	bha,भ	lu,लु	hoo,□

koo,कू	chau,चौ	taa,टा	ne,णे	dham,धं	bhaa,भा	loo,लू	he,हे
ke,के	chan,चं	ti,टि	nai,णै	dhah,धः	bhi,भि	le,ले	hai,है
kai,कै	cham,चं	tee,टी	no,णो	n,न	bhee,भी	lai,लै	ho,हो
ko,को	chah,चः	tu,टु	nau,णौ	na,न	bhu,भु	lo,लो	hau,हौ
kau,कौ	chha,छ	too,टू	nan,णं	naa,ना	bhoo,भू	lau,लौ	han,हं
kan,कं	chhaa,छा	te,टे	nam,णं	ni,नि	bhe,भे	lan,लं	ham,हं
kam,कं	chhi,छि	tai,टै	nah,णः	nee,नी	bhai,भै	lam,लं	hah,हः
kah,कः	chhee,छी	to,टो	t,त	nu,नु	bho,भो	lah,लः	ksh,□
kh,ख	chhu,छु	tau,टौ	ta,ता	noo,नू	bhau,भौ	v,व	ksha,□
kha,ख	chhoo,छू	tan,टं	taa,ता	ne,ने	bhan,भं	va,व	kshaa,□ा
khaa,खा	chhe,छे	tam,टं	ti,ति	nai,नै	bham,भं	vaa,वा	ksha,□ा
khi,□ख	chhai,छै	tah,टः	tee,ती	no,नो	bhah,भः	vi,वि	kshi,ि□
khee,खी	chho,छो	th,ठ	tu,तु	nau,नौ	m,म	vee,वी	kshee,□ी
khu,खु	chhau,छौ	tha,ठा	too,तू	nan,नं	ma,म	vu,वु	kshu,□ु
khoo,खू	chhan,छं	thaa,ठा	te,ते	nam,नं	maa,मा	voo,वू	kshoo,□ू
khe,खे	chham,छं	thi,ठि	tai,तै	nah,नः	mi,मि	ve,वे	kshe,□े
mba,म्ब	dm,□	phi,फ़ि	jam,जं	p,प	mee,मी	gnaa,□ा	
rsh,श□	dr,ड□	phee,फी	jah,जः	pa,प	mu,मु	Gni,ि□	
pra,□	dr,□	phu,फु	z,झ	paa,पा	moo,मू	gnee,□ी	
mru,मृ	dhr,□	phoo,फू	za,झ	pi,पि	me,मे	gnu,□ु	
rva,व□	dv,□	phe,फे	zaa,झा	pee,पी	mai,मै	gnoo,□ू	
kk,□	dy,□	phai,फै	zi,झि	pu,पु	mo,मो	gne,□े	
kka,□ा	ft,□	pho,फो	thee,ठी	poo,पू	mau,मौ	gnai,□ै	
kke,□े	gr,□	phau,फौ	thu,ठु	pe,पे	man,मं	gno,□ो	
kki,□ी	gy,□	my,मी	thoo,ठू	pai,पै	mam,मं	gnau,□ौ	
kkai,□ै	khm,□	khai,खै	the,ठे	po,पो	mah,मः	gnam,□ं	
kku,□ु	khy,□	kho,खो	thai,ठै	pau,पौ	y,य	gnah,□ः	
kru,कृ	kl,□	khau,खौ	tho,ठो	pan,पं	ya,य	bb,□	
gru,गृ	kr,□	khan,खं	thau,ठौ	pam,पं	yaa,या	Bhr,□	
ghru,घृ	ks,□	kham,खं	than,ठं	pah,पः	yi,यि	br,□	
nru,नृ	ky,□	khah,खः	tham,ठं	f,फ	yee,यी	ddh,□	
vru,वृ	ld,□	g,ग	thah,ठः	fa,फ	yu,यु	dhr,ढ□	
shtra,’□	lm,□	ga,गा	d,ड	faa,फा	yoo,यू	tn,□	
shree,□ी	mb,□	gaa,गा	da,ड	fhi,फि	ye,ये	tren,ट□	
shri,□ी	mh,□	gi,गि	daa,डा	fee,फी	yai,यै	tt,□	
kra,□	nd,□	gee,गी	di,डि	vai,वै	kshai,□ै	vr,□	
kraa,□ा	nn,□	gu,गु	dee,डी	vo,वो	ksho,□ो	hw,□	
gra,□ा	ns,□	goo,गू	du,डु	vau,वौ	kshau,□ौ	dhy,□	
rgi,ग□	nt,□	ge,गे	doo,डू	van,वं	kshan,□ं	dhya,□ा	
swa,□	pr,□	gai,गै	de,डे	vam,वं	ksham,□ं	dhyaa,□ा	
sva,□	rc,च□	go,गो	dai,डै	vah,वः	kshah,□ः	w,व	
gve,□े	rf,फ़□	gau,गौ	to,तो	sh,श	dny,□	wa,व	

bda,□	rs,स□	gan,गं	tau,तौ	sha,श	dnya,□ा	waa,वा
hin,हिं	rt,त□	gam,गं	tan,तं	shaa,शा	dnyi,ि□	shw,□
di,दी	rth,थ□	chhah,छः	tam,तं	shi,शि	dnyee,□ी	shwa,□ा
nt,त	ry,य□	j,ज	tah,तः	shee,शी	dnyu,□ु	shwaa,□ा
nd,द	shr,□	ja,ज	tha,थ	shu,शु	dnyoo,□ू	pt,□
shan,शां	shr,□	jaa,जा	thaa,था	shoo,शू	dnye,□े	lk,□
lin,लिं	shv,□	ji,जि	thi,थि	she,शे	dnyai,□ै	ph,फ
ad,अद	sj,□	jee,जी	thee,थी	shai,शै	dnyo,□ो	pha,फ
dwai,□ै	sk,□	ju,जु	thu,थु	sho,शो	dnyau,□ौ	phaa,फा
dve,□े	sn,□	joo,जू	thoo,थू	shau,शौ	dnyan,□ँ	
wai,वै	st,□	je,जे	the,थे	shan,शं	dnyam,□ँ	
bhon,भों	st,□	jai,जै	thai,थै	sham,शं	dnyah,□ः	
sw,□	sth,□थ	jo,जो	tho,थो	tham,थं	chch,□	
xn,ङ	sv,□	jau,जौ	thau,थौ	thah,थः	chchh,□	
gna,□	thr,□	jan,जं	than,थं	d,द	chhu,□	

ANNEXURE E: Hindi to English conversion list (Maatra for Transliteration process)

Roman	Hindi Maatra
a	ा
a	ा
i	ि
ii	ी
u	ु
uu	ू
r	ृ
e	े
ai	ै
o	ो
au	ौ
n	ँ
m	ं

ANNEXURE F: Sample of Hindi to English dictionary of common words

English	Hindi	English	Hindi	English	Hindi	English	Hindi
aah	आह	aanevaala	आनेवाला	baadha	बाधा	chalte	चलते
aaha	आहा	aanevaalaa	आनेवाला	baadi	बाड़ी	chaltey	च□े
aahaa	आहा	aanevaale	आनेवाले	baadlon	बादलों	chalte	छलते
aahaahaa	आहाहा	aanevaalii	आनेवाली	baadshaah	बादशाह	chalthi	चलती
aahat	आहत	aanewaala	आनेवाल	baadshaah	बाद् शाह	chalti	चलती
aahaten	आहट□	aanewaala	आनेवाला	baadshaahon	बाद् शाहों	chalu	चालू
aahaton	आहटों	aanewaale	आनेवाले	baadshah	बादशाह	chalun	चलूँ
aahe	आहे	aanewaali	आनेविाल	baadshahon	बादशाहों	chalun	चलूं
aahee	आहे	aanewala	आनेवाला	baaen	बाएं	chalungee	चलूँगी
aahen	आह□	aanewale	आनेवाले	baag	बाग	chalungi	चलूँगी
aahista	िआह□	aanewaley	आनेवाले	baag	बाग़	chaluun	चलूँ
aahistaa	िआह□	aang	अंग	baag	भाग	chaluun	चलूं
aaho	आहो	aangan	आँगन	baaga	बागा	chaluunga	चलूंगा
aahon	आहों	aanganaa	आँगना	baagad	बागड़	chaluungaa	चलूंगा
aaida	आईदा	aanganon	आँगनों	baageshvari	बागे□री	chaluungi	चलूँगी
aaika	आईका	aangdayi	अंगड़ाई	baaghi	बागी	chalye	चले
aaina	आइना	aangna	अंगना	baaghon	बागों	cham	चम
aaina	आईना	aangna	आँगना	baaghon	बाग़ों	cham	छम
aainaa	आइना	aanhe	आँहे	baago	बागों	chama	चम
aainaa	आईना	aanhen	आँह□	baagon	बागों	chama	छमा
aaine	आइने	aanhon	आँहों	baagon	बाग़ों	chamaacham	चमाचम
aainey	आईने	aankade	आँकड़े	baagon	बाहों	chamacha	चमचा
aais	आइस	aankana	आँकना	baah	बाँह	chamacham	चमचम
aaitabaar	आइतबार	aankein	आँख□	baahaake	बाहाके	chamache	चमचे
aaitbaar	आइतबार	aankh	आँख	baahaane	बाहाने	chamak	चमक
aaiya	आइया	aankhe	आँखे	baahaar	बाहार	chamak	छमक
aaiye	आइये	aankhein	आँख□	baahar	बहार	chamaka	चमका
aaiyey	आईये	aankhen	आँख□	baahar	बाहर	chamakaa	चमका
aaiyega	आइयेगा	aankhen	आँखेन	baaharakii	बाहरकी	chamakaae	चमकाए
aaiyo	आइयो	aankhey	आँख□	baahe	बहे	chamakaakar	चमकाकर
aaj	आज	aankhiyaan	अँ□खयाँ	baahe	बाँह□	chamakaana	चमकाना
aaja	आजा	aankhiyaan	अं□खयाँ	baahe	बाहे	chamakaao	चमकाओ
aajaa	आजा	aankho	आँखो	baahe	बाह□	chamakaati	चमकाती
aajaae	आजाए	aankho	आँखों	baahein	बाँह□	chamakaauun	चमकाऊं
aajaaegaa	आजाएगा	aankhoen	आँखोएन	baahein	बाह□	chamakaaye	चमकाये
aajaaein	आजाएं	aankhon	आँखों	baahen	बाँह□	chamakaayen	चमकाय□
aajaao	आजाओ	aankkhon	आँखों	baahen	बाह□	chamakaayi	चमकाई
aajaate	आजाते	aankon	आँखों	baahen	बाहों	chamakaayu□	चमकाऊँ
aajaaye	आजाये	aanso	आँसू	baahir	बिाहर	chamakan	चमकन
aajaayen	आजाएं	aansoo	आँसू	baaho	बाँहों	chamakana	चमकना
aajama	आजमा	aansu	आँसु	baaho	बाहों	chamakane	चमकन

ANNEXURE G: Transliteration in example sentences with output

No.	Input (Romanized Hindi)	Output (Transliterated Text)
1	Kya aapne humare lia khana laya hai? Aj meri maa ne dabba nai dia hai humko	□ा/H आपने/H हमारे/H िलए/H खाना/H लाया/H ह□/H अज/H मे□र/H मां/H any/Eद□ा/H नै/H िदया/H ह□/H हमको/H
2	Aaj kal garmi kitni badh gai hai. Ghar se bahar nikalne ke lia bohot taklif hoti hai.	आज/H कल/H गरमी/H िक□ी/H बढ़/H गयी/H ह□/H घर/H से/H बाहर/H िनकलने/H के/H िलए/H ब□त/H ता□□फ/Hहोतीं/H ह□/H
3	Baki sabko bhi bata dena, aaj ka khana mere ghar pe hai. Mere naya ghar dekhne zaroor ana tum sab log.	बाकी/H सबको/H भी/H बता/H देना/H आज/H का/H खाना/H मेरे/H घर/H पे/H ह□/H मेरा/H नया/H घर/H देखने/H ज़□र/H आना/H तूम/H सब/H log/E
4	Kya hum aaj ghumne jaa sakte hai. Aaj meri chhutti hai.	□ा/H hum/Eआज/H घुमने/H जा/H सकते/H ह□/H आज/H मे□र/H छु ि□/H ह□/H
5	Vo kya hai na, aaj meri dost ki shaadi hai, toh isiliye mai itne saj daj ke nikli hu ghar se.	वो/H क्या/H है/H ना/H आज/H मेर/H dost/E िक/H शादी/H ह□/H तोह/H इसीलए/H म/H इतने/H सज/H दज/H के/H िनकली/H □/H घर/H से/H
6	Mere ghar walon ne mujhe aj raat tere ghar pe rukhne k lia kaha hai. Mere mata aur pita bahar jaa rahe hai na, isiliye.	मेरे/H घर/H वालों/H any/Eमुझे/H आज/H रात/H तेरे/H घर/H पे/H □खने/H ok/Eिलए/H कहा/H ह□/H मेरे/H माता/H और/H िपता/H बाहर/H जा/H राह□/H ह□/H ना/H इसीलए/H
7	Mujhe kab milaoge apne doston se, me sirf unki kahaniyan sun rahi hu.	मुझे/H kab/Eिमलाओगे/H अपने/H दो□ों/H से/H me/Eिसफ़□/H उनकी/H कहािनयाँ/H sun/Eरही/H □/H
8	Mujhe ek din ki chhutti chahiye, jab mai poori din kitaab lekar ghar pe baithke padh sakti hu.	मुझे/H एक/H िदन/H िक/H छु ि□/H चिहये/H जब/H म□/H पूरी/H िदन/H िख़ताब/Hलेकर/H घर/H पे/H बैठके/H पढ़/H सकती/H □/H
9	Aj bohot thak gai hu yaar, kuch paani ya kuch de dena.	आज/H बत/H थक/H गयी/H □/H यार/H कु छ/H पाणी/H ya/E कु छ/H दे/H देना/H
10	Mujhe tum paise kab tak vapis dogi, tangi padhi hai yaar. Thoda jaldi se de dena, mere bhi halat ho raha hai.	मुझे/H तूम/H paise/E कब/H तक/H विपस/H दोगी/H तंिग/H पढ़ी/H ह□/H यार/H थोड़ा/H ज□ी/H से/H दे/H देना/H मेरे/H भी/H हालत/H हो/H रहा/H ह□/H

ANNEXURE H: Wordplay in example sentences with output

No.	Input (with elongated words)	Output (normalized text)
1	Heyyyy! I am sooo glad I ran into you-uuu! Common lets go outttttttt	hey/E i/E am/E so/E glad/E i/E ran/E into/E you/E common/E lets/E go/E out/E
2	YOOO! Are you coming to the par-tyyyyyy! I heard its going to be a blasssst	yo/E are/E you/E coming/E to/E the/E party/E i/E heard/E it/E is/E going/E to/E be/E a/E blast/E
3	Omgggggggg! Did you listen to that new Ed Sheeran songgggggg!! Its out of thisssss worlddddd.	oh/E my/E god/E did/E you/E listen/E to/E that/E new/E ed/E शीरं/H song/E it/E is/E out/E of/E this/E world/E
4	The new hippie lifestyle is sooooo cooool! I want to try itttttt sooooo bad-lyyyy mannnn	The/E new/E hippie/E lifestyle/E is/E so/E cool/E i/E want/E to/E try/E it/E so/E badly/E man/E
5	Heyyyy... will you dance at my weddinggggggg!! Pleaseeeee yaaaarrr... Dancers are neeeeeded dude!	hey/E will/E you/E dance/E at/E my/E wedding/E please/E यार/H dancers/E are/E needed/E dude/E
6	Like, did you seeeee the new Trump rul-ingssssss, like what was he thinking li-keeee	like/E did/E you/E see/E the/E new/E trump/E rulings/E like/E what/E was/E he/E thinking/E like/E
7	The lil flea market is sooooo cooolll.. I reallllyyyy like the vibes that place gives dudeeeee	the/E little/E flea/E market/E is/E so/E □□□□□□□□ल/H i/E रे□□□□/H like/E the/E vibes/E that/E place/E gives/E dude/E
8	Did you seeee the Disney movieeee!! It's a book based movie on the life of Belle!!!! Its awsmmmmmm	did/E you/E see/E the/E disney/E movie/E it/E s/E a/E book/E based/E movie/E on/E the/E life/E of/E belle/E it/E is/E awesome/E
9	Watchhhhh anddddd learnnnnnn!!!!! That's howwwwwww itsssss doneeee...	watch/E and/E learn/E that/E s/E how/E it/E is/E done/E
10	I ammmmm sooooo madddddd at uu-uuu rtttttt nwwwwww.. get outttttttt of hereeeeee	i/E am/E so/E mad/E at/E you/E right/E now/E get/E out/E of/E here/E

ANNEXURE I: Slang example sentences with output

No.	Input (with slang)	Output (normalized text)
1	Hey cn u ack the receipt of the product at ur end	hey/E can/E you/E acknowledge/E the/E receipt/E of/E the/E product/E your/E end/E
2	Reply asap. The mtng is imp nd I want u thr	reply/E as/E soon/E as/E possible/E the/E meeting/E is/E important/E and/E i/E want/E you/E there/E
3	Cn u b ne more annoying, she said. His reply ws masked in the silence of his sadness nd agony	can/E you/E be/E any/E more/E annoying/E she/E said/E his/E reply/E was/E masked/E in/E the/E silence/E of/E his/E sadness/E and/E agony/E
4	BRB gotta eat smt M famished r8 nw	be/E right/E back/E gotta/E eat/E something/E i/E am/E famished/E right/E now/E
5	Cmb urgent wrk here now HRU nd hwz ur health	call/E me/E back/E urgent/E work/E here/E now/E how/E are/E you/E and/E how/E is/E your/E health/E
6	Hey Elon Musk started a new startup! I m gonna apply thr dude. It's a once in a lifetime opportunity	hey/E एलों/H musk/E started/E a/E new/E startup/E i/E i/E am/E going/E to/E apply/E there/E dude/E it/E s/E a/E once/E in/E a/E lifetime/E
7	So hws life treating u R u happy wid d way things are proceeding r8 nw	so/E □स/H life/E treating/E you/E are/E you/E happy/E with/E the/E way/E things/E are/E proceeding/E right/E now/E
8	Any1 intererested in buying this d2d use chair from us, its in f9 condition	anyone/E interested/E in/E buying/E this/E day/E to/E day/E use/E chair/E from/E us/E it/E is/E in/E fine/E condition/E
9	Hrt diseases kills many Indians every year	heart/E diseases/E kills/E many/E indians/E every/E year/E
10	n/t is allowed during exams	and/E t/E is/E allowed/E during/E exams/E

ANNEXURE J: Abbreviations in example sentences with output

No.	Input (with abbreviations)	Output (expanded text)
1	Dmi I am glad to help you df	don't/E mention/E it/E i/E am/E glad/E to/E help/E you/E dear/E friend/E
2	Dwb its an automatic reply	don't/E write/E back/E it/E is/E an/E automatic/E reply/E
3	After watching friends icl its rofl nd lol funny man!	after/E watching/E friends/E इ□□□□/H it/E is/E rolling/E on/E the/E floor/E laughing/E and/E laugh/E out/E loud/E funny/E man/E
4	Abd work needs to be add by the owner of the works, kindly comply	already/E been/E done/E work/E needs/E to/E be/E add/E bye/E the/E owner/E of/E the/E works/E kindly/E comply/E
5	Bb , hand and call me when you reach at your destination	bye/E bye/E have/E a/E nice/E day/E and/E call/E me/E when/E you/E reach/E at/E your/E destination/E
6	Cpg is highly overrated, one should use organic items instead	consumer/E packaged/E goods/E is/E highly/E overrated/E one/E should/E use/E organic/E items/E instead/E
7	Cot is the secret for every successful relationship in this world	circle/E of/E trust/E is/E the/E secret/E for/E every/E successful/E relationship/E in/E this/E world/E
8	Dmi , it was my pleasure	don't/E mention/E it/E it/E was/E my/E pleasure/E
9	Fb is my fav social media / social networking site man!	फ□□□□□□□□बूक is/E my/E favorite/E social/E media/E social/E networking/E site/E man/E
10	Gmab , is what ross was intending to tell Rachel, but it came out all wrong.	give/E me/E a/E break/E is/E what/E ross/E was/E intending/E to/E tell/E राचेल/H but/E it/E came/E out/E all/E wrong/E